Applying Automotive EDR Data to Traffic Crash Reconstruction

Introduction

Presented by Rick Ruth
313 910 5809
ruthconsulting@comcast.net

in (Ctrl+Plus)

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Introduction -Main Messages

1. EDR's improve recon **accuracy** & get to **causation** – use with **all other** available scene evidence – but if other evidence is lacking, EDR can be used a PRIMARY EVIDENCE.
2. The EDR strength is to give us speed **PRIOR** TO *BRAKING* OR *LOSS OF CONTROL, &* DRIVER BEHAVIOR on gas and brake pedal. Modern ABS brakes TOOK AWAY your tire marks, only EDR can "give you back" speed prior to braking. Newer data elements give us EVEN MORE INSIGHT into precrash behavior.
3. Event Data Recording capability and data accessibility ***varies widely*** by **manufacturer, model *and* model year.** Coverage keeps growing with MORE AND BETTER DATA available with each new model released.

Main Messages

4. **Minimum** Data in **new** US vehicles equipped with EDR was ***STANDARDIZED*** Sept 2012 by NHTSA Part 563 rule, ensures public access to data, and that data survives in severe crashes.
5. Proposed rule to REQUIRE EDR on 100% of US passenger vehicles by 9/2014 was *NOT implemented!* *Coming ..ever?*
6. EDR's will NEVER replace human beings in figuring out what really happened.
7. PRIVACY – New US law Dec 2015 requiring owner permission or a warrant.

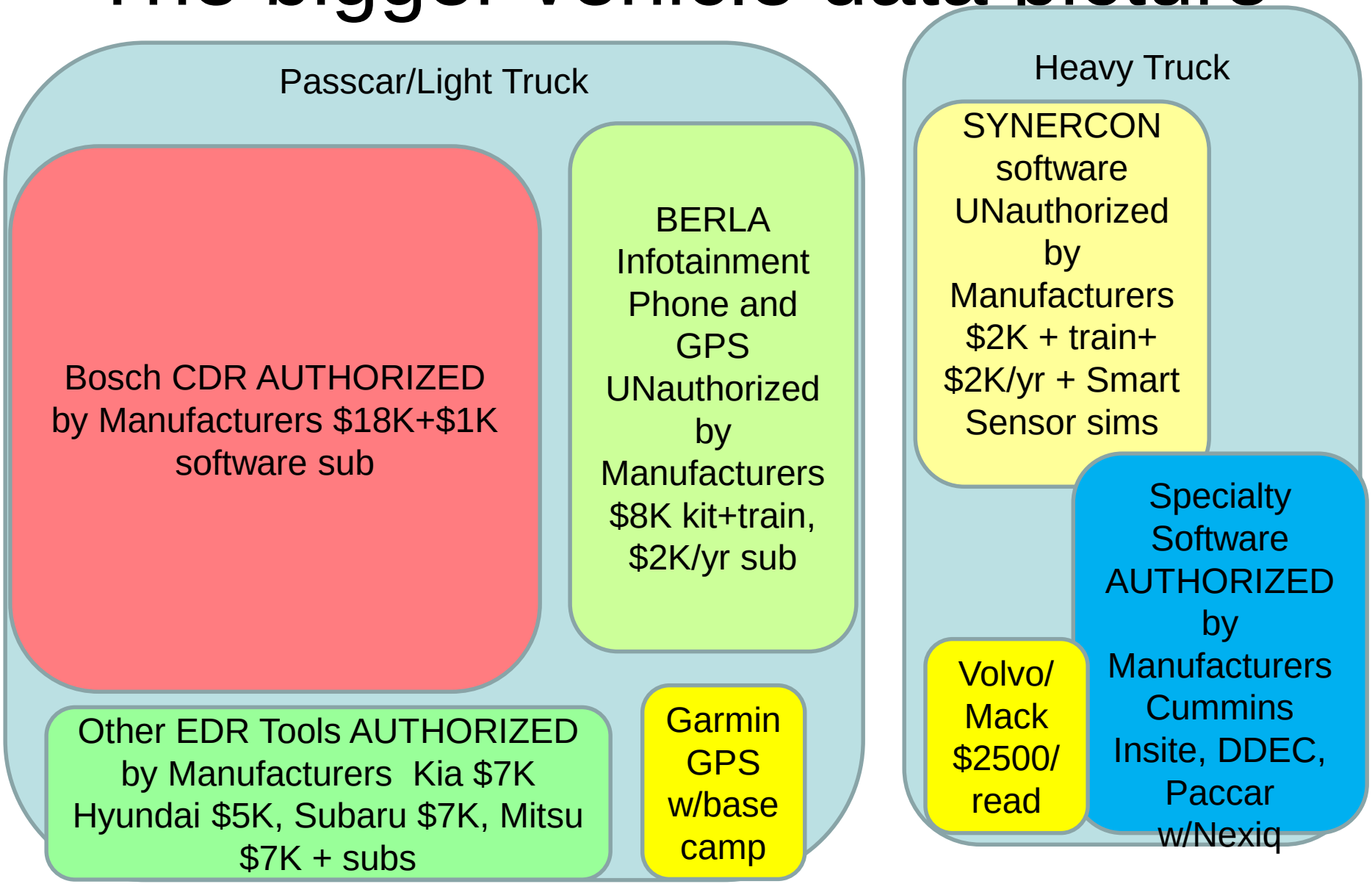
Keeping EDR's in Perspective

- There are 289 million vehicles registered in the US & CA (93% US, 7% Canada), **54%** have an EDR with data accessible by the Bosch CDR system
- 88+% of 2017's are supported by Bosch.
- Another 11% have EDR accessible by other tools. **99%** have EDR.
- In two car collisions, on average one will have a CDR supported EDR. With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles.

Scope of EDR Investigators

- 1800 CDR Software subscriptions.
- Over half Law Enforcement 54%
- Private reconstructionists 41%
- Insurance 1.5%, Govt, Fleets rest
- New EU reg coming like 563
- Europe and Australia – mostly Toyota, Volvo, FCA, Holden – limited coverage – .
- New Korean reg Dec 2015.
- Tokyo police 50 kits. China next.

The bigger vehicle data picture

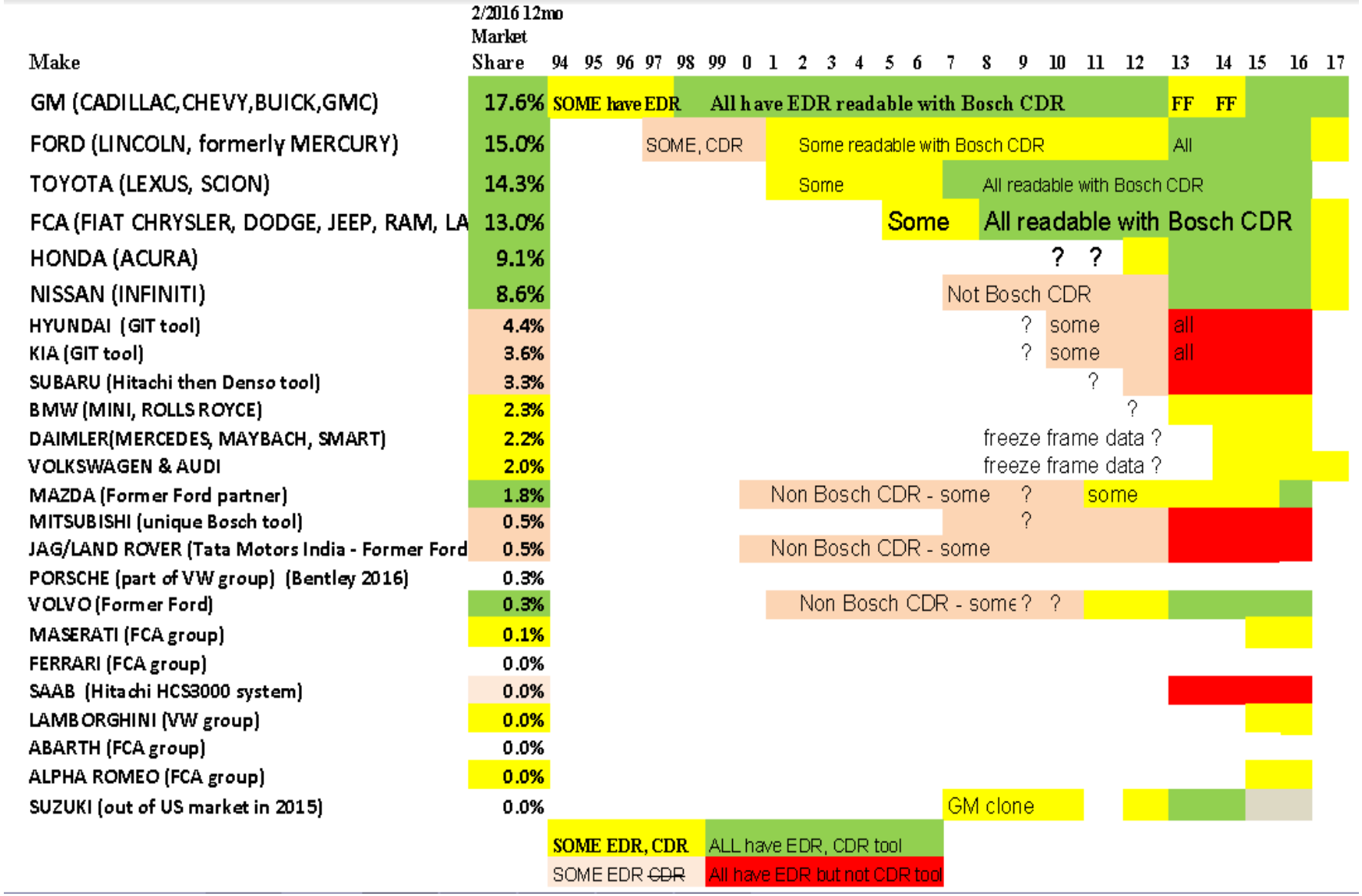


US/CA Lt. Veh EDR & Tool Used

Manufacturer	12 months Sales end 2/16	Market Share	2016 EDR?	Tool?
GM	3,079,772	17.6%	YES	Bosch CDR
Ford	2,634,491	15.0%	YES	Bosch CDR
Toyota	2,498,889	14.3%	YES	Bosch CDR
Chrysler Group	2,273,230	13.0%	YES	Bosch CDR
Honda	1,598,383	9.1%	YES	Bosch CDR
Nissan	1,499,020	8.6%	YES	Bosch CDR
Hyundai	762,720	4.4%	YES	GIT tool co.
Kia	631,531	3.6%	YES	GIT tool co.
Subaru	583,617	3.3%	YES	Denso 2016+, was Hitachi
BMW	400,064	2.3%	2015	Bosch CDR
Mercedes-Benz	380,690	2.2%	2014	Bosch CDR
Volkswagen	342,626	2.0%	2015	Bosch CDR
Mazda	314,510	1.8%	YES	Bosch CDR
Audi	202,774	1.2%	2015	Bosch CDR
Mitsubishi	95,450	0.5%	YES	OTC/Bosch SPX
Tata (Jag, LR)	87,120	0.5%	YES	OTC/Bosch SPX
Geely (Volvo)	71,744	0.4%	YES	Bosch CDR
Porsche	52,532	0.3%	NO	N/A
Industry Total	17,509,163	100.0%	99.3%	Bosch CDR 87.0%
updated April 4 2016				Non CDR 12.3%

EDR Phase-in by Mfr

Data is incomplete for smaller mfrs with no public tool



2017 Recording Capability*

*based on 2012

	Part 563 Table 1					Part 563 Table 2						Beyond 563								
Mfr	Event Complete	Key Cycles	Long. ΔV	Speed/Brake/ Throttle or Accel	Multi Event	Lateral ΔV	RPM	Steering Angle	ACM Acceleration	ABS on/off	Roll Angle	ESC Long Accel	ESC Lat Accel	Yaw Rate	Tire Pressure	Panic Brake Assist	Wheel Speeds	Roll Rate	Brake Pressure	Day/Date/Time of Crash
GM	X	X	X	5@0.5 TA	X	X	X	-	X, Y, Z 300@2	X		-			LO	-	-	1 @ 10ms		
Ford	X	X	X	5@0.5 A	X	X	X	5@0.1	-	X		5@0.1		5@0.1	LO		-	5@0.1		
Chrysler	X	X	X	5@0.1 TA	X	X	X	5@0.1	-	X		-		5@0.1	PSI	X	X			
Toyota	X	X	X	5@0.5 TA	X	Side	X	5@0.5	-	?	?	5		5@0.5	-	-	-	In Roll	5@0.5	
Honda	X	X	X	5@0.5 TA	X	X	X	5@0.5	X, Y, Z 250@10	X	-1+5@0.1	-		-	-	-	-			
Nissan	X	X	X	5@0.5 A	X	X	X	5@0.5	X, Y 250@10	-	-	-		-	-	-	?			
Mazda	X	X	X	5@0.5 T	X	X	X	5@0.5	X, Y 250@10	-							-			
Volvo	X	X	X	5@0.5T	X	-	-	-	X,Y,Z 250@10	-	-1+0.3@0.1									
BMW	X	X	X	5@0.5A	X		X	5@0.5	X, Y 300@10ms	X	-									
Mercedes	X	X	X	5@0.5A	X															
VW	X	X	X	5@0.5A	X	X														
Hyundai/Kia	X	X	X	5@0.5	X	X	X	5@0.5	X,Y,Z 250@10	X	-1+5@0.1	-		-	-	-	-			
Subaru	X	X	X	5@0.5A	X	X	X	5@0.5		X										
Mitsubishi	X	X	X	5@0.5A	X	X	X		X, Y 250@10											

Bosch CDR Hardware Update

- *No **functional** change* to **interface** module since ‘05 -Blue CAN Plus, Green, and new Green “CE” (like UL) are all OK
- Bosch has managed new technology with adapters
- New diagnostic protocols like CAN-HD and Ethernet have driven a new interface module, the CDR-900. On sale by year end for under $4,000.

Bosch CDR 900

The Next Generation Crash Data Retrieval Tool

Coming soon for North America



Planned for release by the end of 2017

The **Bosch CDR 900** is a high-performance vehicle communications interface to retrieve crash data stored in a vehicles event data recorder (EDR) after a crash.

Introductory pricing starting below $4,000 USD

CDR900

- Initially for NEW vehicles
- Will eventually handle legacy vehicles too…..

Transition back to single tool

Frequently Asked Questions

What is the Bosch CDR 900?

The Bosch CDR 900 is a high-performance Vehicle Communications Interface (VCI) device used to connect and communicate with many complex vehicle diagnostic networks through the OBD connector and communicate directly with vehicle electronic control modules (ECUs). When connected to a Windows-based personal computer running the Bosch CDR Tool software, the Bosch CDR 900 is used for retrieval of crash data stored in one or more vehicles ECUs such as airbag control modules and pedestrian protection ECUs. The Bosch CDR 900 employs the lasted technology for communications with vehicle networks supporting the latest vehicle communications protocols.

Will the Bosch CDR 900 replace the existing Bosch CDR CAN*plus* Vehicle Interface Module? If so, when?

*Yes, but not until sometime **after** 2019. In the meantime, the Bosch CDR Tool user will need to own both the Bosch CDR 900 and the Bosch CDR CANplus Module to access all vehicles listed in the CDR help file.*



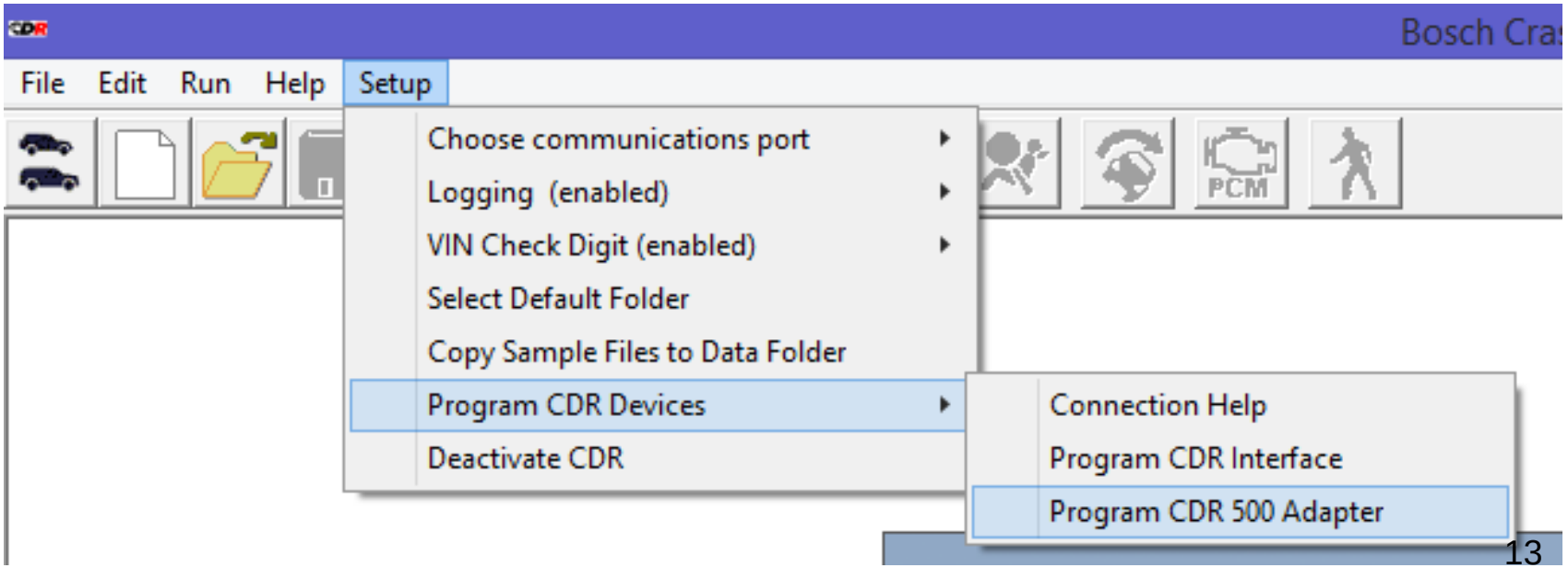
End of 2017 After 2019

Why is the Bosch CDR 900 needed?

*The Bosch CDR CANplus Module technology has been around for 17+ years with a hardware refresh occurring 12 years ago. Although the current CDR CANplus Module continues to be a reliable interface for retrieval of EDR data, new technologies cannot be supported in a **refresh** of the current hardware. New technologies include CAN FD, DoIP, faster communication speeds and new PC interface technologies to name a few . To continue with the rapid evolution of EDR data collection, Bosch is moving forward with a new vehicle communications interface to handle these new technological demands.*

(Relatively) New Adapters

- **CDR 500 NOT NEEDED for DLC READOUTS-vehicle gateway module performs CDR 500 function**
- **CDR 500 “Flex Ray” direct to module adapter for BMW hybrids & some VW/Audi, more to come, cost $2500**
- **ACM only communicates in flex ray, the CDR 500 translates from vehicle CAN to Flex Ray.**
- **Flex Ray module requires firmware updates now with newer software (see below)**



More Hardware Updates

- Proliferation of **DLC adapters** as manufacturers use more than 1 can bus (low speed, medium speed, high speed) and Bosch must connect to nonstandard pins in the DLC to find the correct CAN bus for the ACM (Ford, Fiat, Toyota)
- New ACM's have big capacitors with big inrush currents to charge them. Must use #2435 2.5 amp current limited power supply for 387 "power booster" adapter or risk damaging it

HARDWARE CHECKLIST 17.5

1699200261	Cable, ID816 CDR Volvo ACM	16.2
1699200264	Cable, ID819 CDR FCA ACM	16.2
1699200369	Adapter, DLC 3	16.2
1699200129	Cable: ID813 CDR Audi ACM (requires CDR 500 Adapter)	16.3
1699200143	Cable, ID814 CDR Toyota ACM	16.3
1699200144	Cable, ID815 CDR Toyota ACM	16.3
1699200262	Cable, ID817 CDR Toyota ACM	16.3
1699200263	Cable, ID818 CDR Toyota ACM	16.3
1699200365	Cable, ID821 CDR GM ACM	16.3
1699200953	Cable, USB to Serial, 6', DB9 Male	16.4
1699200428	Cable, ID823 CDR Ford ACM	16.5
1699200383	Cable, ID822 CDR BMW ACM 5 (requires CDR 500 Adapter)	17.0
1699200440	Cable, ID824 CDR Karma ACM	17.0
1699200466	Cable, ID825 CDR Honda ACM	17.2
1699200510	Adapter, DLC 4 (used for Toyota	17.2
1699200559	Cable, ID827 CDR Toyota ACM	17.2

New 3-digit Part ID,

2015: Toyota Cables for non US markets released. Not required in US.

Jan 2017

Not required in US.

EMERGING TRENDS

- More manufacturers are becoming concerned about cyber security, of hackers trying to take over vehicle control systems.
- Many are installing “gate-way” modules or features to protect ECU’s from hacking.
- For CDR DLC users, “back powering” may not work on newer vehicles by powering the ACM fuse alone. Direct to module may become required on some.

Recent CDR Software Updates

- 16.5 Mar. 18, 2016- GM/Ford/Honda 2017's
- 16.6 May 23, 2016 – FCA/Volvo/VW 2017's
- 17.0 August 26, 2016 – License holder ID, more 2017's, new Karma brand, Korea/China
- 17.1 Nov. 2, 2016 – more 2017 models
- 17.2 Jan 26, 2017 – GM 2018's!
- 17.3 April 11, 2017 – More 2018's
- 17.4 June 15, 2017 – More 2018's
- 17.4.1 & 17.4.2 – fixes, some more 2018's
- 17.5 Oct 17, 2017 – more 2018's

The 17.5 Story

- Every year we get a new model year of coverage, and every year another year of old cars without EDR's get off the road. The covered vehicles on the road with CDR is 54% and climbing about 4%/yr. The total available from manufacturers and other tools is much more!!
- Canada benefits from the USA EDR reg. Europe, Australia, Korea, and China are just getting started!

Other new software features

- “Navigation pane” added on left in bigger FCA reports to let you jump to section you want to.
- Auto com port selection added (hooray!)
- CDR optimizes which printer driver makes PDF files (but Windows 10 may override).

What's Coming in next 12 mo?

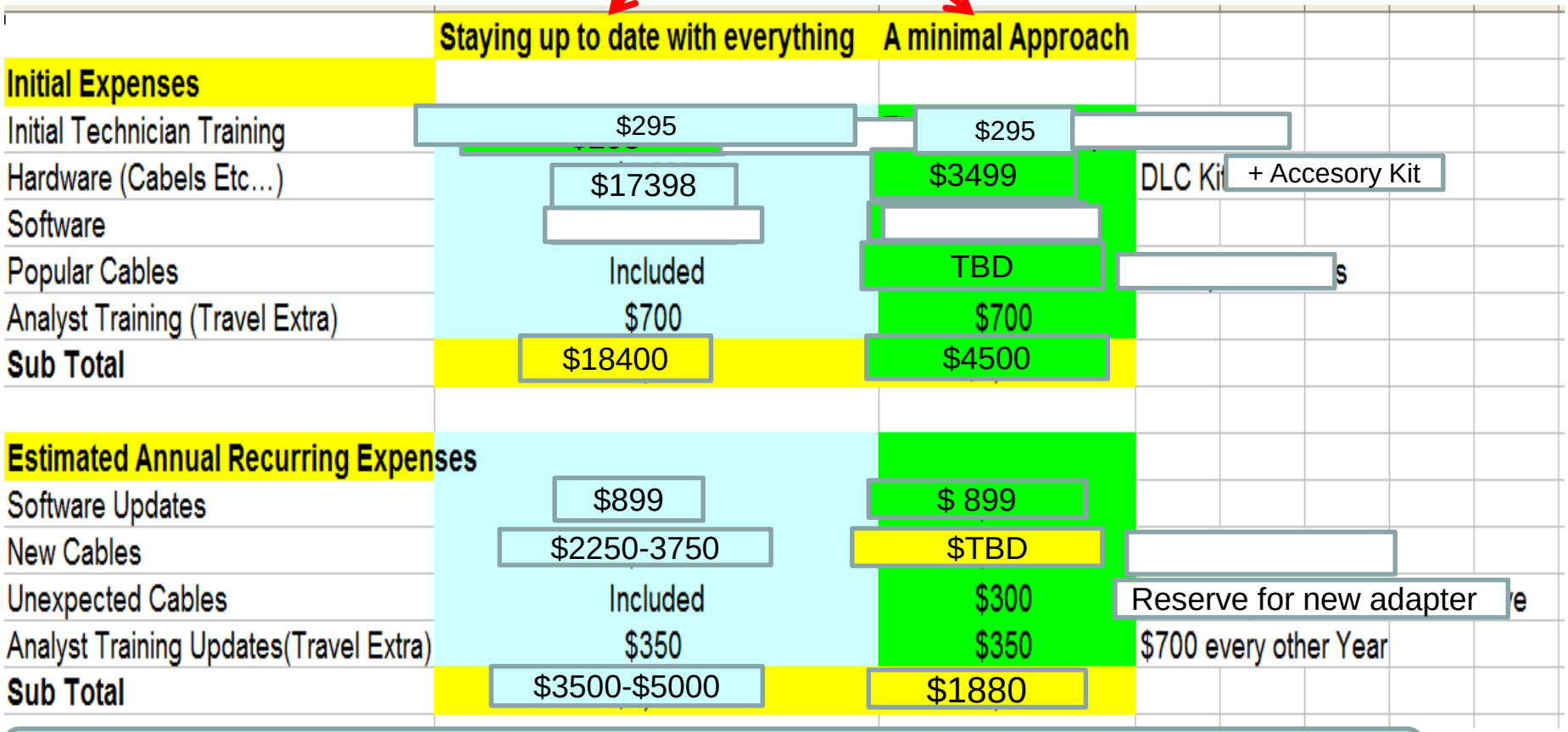
- One software release per quarter
- Another 5 or 6 cables in 2017 (now 2018?)
- Additional manufacturer(s) – Subaru???

  (no timeframe yet)

CDR Budgeting

Two approaches to a budget

Add new interface $4,000 to this



	Staying up to date with everything	A minimal Approach	
Initial Expenses			
Initial Technician Training	$295	$295	
Hardware (Cabels Etc...)	$17398	$3499	DLC Kit + Accesory Kit
Software			
Popular Cables	Included	TBD	s
Analyst Training (Travel Extra)	$700	$700	
Sub Total	$18400	$4500	
Estimated Annual Recurring Expenses			
Software Updates	$899	$ 899	
New Cables	$2250-3750	$TBD	
Unexpected Cables	Included	$300	Reserve for new adapter e
Analyst Training Updates(Travel Extra)	$350	$350	$700 every other Year
Sub Total	$3500-$5000	$1880	

Third approach – New CDR Deluxe – covers 80% of cars on road for $9600

New Pedestrian Protection Modules with Data



- Air chambers with pressure sensors in bumper trigger hood pop up feature
- Systems are intended to NOT trigger in car to car crashes (but may)
- Get EDR-style data at less than 5mph/8kph normal recording threshold thanks to the xtra sensors!!
- FCA, GM, Toyota actively adding them

New Toyota Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for
EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli
list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

This Started in Nov 2015

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this
the help file.

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n
indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

Vehicles Are Listed Alphabetically
Not by Model Year

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

4Runner (Toyota) **2003 and later** **Market: 7**

Year	EDR Generation	CDR Cable & Adapter	Module	Module Location
2002 - 2008	Gen 1 (02EDR)	F00K108617 (no adapter)	ACM	Under Center Stack
2009 - 2011	Gen 2 (06EDR)	F00K108616 & F00K108387	ACM	Under Center Stack
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	ACM	Under Center Stack

Help file may be CALENDAR year not model year,
EDR may not yet be present in Model Year of first year listed

New Toyota Vehicle Lookup

Camry (Toyota) **2001 and later** **Market: 7**

All Markets the Vehicle is Sold

Year	EDR Generation	CDR Cable & Adapter	M
2001 - 2004 2005 (US/Canada)	Gen 1 (00EDR/02EDR)	F00K108615 (no adapter) F00K108617 (no adapter) Click here [?] to match the ACM part number to the correct cable	A
2005 - 2011	Gen 2 (04EDR/06EDR)	F00K108613 (no adapter) F00K108614 & F00K108387 F00K108615 (no adapter) F00K108616 & F00K108387 Click here [?] to match the ACM part number to the correct cable	A
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	A

< Models

Note CAMRY got EDR in 02 Model Year in US

Note CAMRY got 04 EDR in 07 Model Year in US

Note CAMRY got 12 EDR as running change in 12 model year, early models have 04/06 EDR in US

CDR/EDR Training

- Bosch Training Provider Page lists IPTM (Rick Ruth+), Crash Data Specialists (Brad Muir+), CSI (Rusty Haight), CDR-training.com (CSI tech 1&2 trainers) and Northwestern (thru CSI). SAE (Rick R) not listed, teaches analysis only.
- There is no “certification” – your qualifications depend on your training and experience
- Bosch does not “certify” any user or endorse any training organization as “preferred”.
- IPTM has ONLINE technician training $295
- All trainers cover basics, but each has unique style and additional content -try them all!

Additional EDR training available

- IPTM has a “Level 2” class which goes into greater depth on how to use the law of sines and cosines to solve Velocity Vector triangles, offset collisions w/EMR, ground forces & new data elements.
- IPTM EDR Analyst training developed for online – may go live “Q4 2017/Q1 2018”

TIMING OF LAST DATA POINT

- The last precrash data point is recorded sometime between one interval between data points before AE to AE (For data points every 0.5 sec, somewhere between -0.5 sec to AE and AE.
- The Airbag control module cannot predict the future
- For ACM's using a wakeup system, the system does not wake up at first bumper touch. It wakes up when the accelerometer mounted remotely measures a value higher than normal driving, typically 1-2g.
- ***RELATIVELY NEW SUBTLETY***: It may be POSSIBLE to record a value labeled as "pre-crash" between when the bumpers first touch and when the algorithm enables.
- For most practical purposes this can be ignored because the odds of it are small and the effect on the last data point would be small. Analysts may need to quantify the odds or how much the effect could be.

TIME FROM BUMPER TOUCH TO AE

The Halversine pulse waveform is used to represent the “ideal” shape of a crash pulse (no underdamping of sensors)

The Halversine function can be used to calculate the rise time to reach a 2G algorithm wake up for various severities of crash pulses.

Max Pulse g	Time to 2G in ms
<2g	never
2G	50
3G	30
4G	25
5G	21.8
10G	14.7
20G	10.3
30G	8.3
40G	7.2
50G	6.4
60G	5.8
70G	5.4

ODDS OF DATA BEING COLLECTED BETWEEN FIRST BUMPER TOUCH AND AE

		Odds to 1 of Data from start to AE for different intervals between data points		
Max Pulse g	Time to 2G in ms	1 sec	0.5 sec	0.1 sec
<2g	never	NA	NA	NA
2G	50	20	10	2
3G	30	33	17	3
4G	25	40	20	4
5G	21.8	46	23	5
10G	14.7	68	34	7
20G	10.3	97	49	10
30G	8.3	120	60	12
40G	7.2	139	69	14
50G	6.4	156	78	16
60G	5.8	172	86	17
70G	5.4	185	93	19

The stronger the crash pulse the less likely that Pre-Crash data will be collected between first bumper touch and AE.

The longer the interval between Pre-Crash data points the less likely that data will be collected between first bumper touch and AE.

MAGNITUDE OF SPEED LOSS BETWEEN FIRST BUMPER TOUCH AND AE

- The amount of change in velocity is -0.1 mph in a 50G pulse and increases as severity decreases to -0.52 mph in a 3G pulse.
- For non wakeup systems, 563 defines start of event as where speed change is 0.5mph over 20ms.
- These effects are small and can be ignored in most high speed collisions. In low speed collisions it may be a larger percentage of the total change in velocity.

		Odds to 1 of Data from start to AE for different intervals between data points			Delta V
Max Pulse g	Time to 2G in ms	1 sec	0.5 sec	0.1 sec	MPH
<2g	never	NA	NA	NA	
2G	50	20	10	2	
3G	30	33	17	3	-0.52
4G	25	40	20	4	
5G	21.8	46	23	5	-0.35
10G	14.7	68	34	7	-0.23
20G	10.3	97	49	10	-0.16
30G	8.3	120	60	12	
40G	7.2	139	69	14	
50G	6.4	156	78	16	-0.1
60G	5.8	172	86	17	
70G	5.4	185	93	19	

Inertial Brake Pedal Application

- The question is whether the EDR could report the brake on at the last data point due to inertial brake pedal application.
- Theoretically, if a crash were severe enough, inertial forces on the brake pedal could become sufficient to overcome friction and stroke the brake pedal enough to activate the brake light switch reported in EDR's.
- There has not been testing in this area – it is not well known or understood if this would take 30G, 50G, 70G, etc. and could depend on vehicle design.
- Precrash data should cut off at AE when forces are 2G or less, which would NOT be sufficient to stroke the brake pedal. Any stroking would take place much later in the crash pulse. It should not be reflected in precrash data.
- However, in staged collisions of GM SDM-G vehicles (early 2000's design), a small number of EDR files have reported brake pedal as "on" during the last data point. These remain unexplained as of this writing. Most staged collisions do NOT report the brake pedal as on in the last data point.
- **Videos of some NHTSA crash tests show brake lights may illuminate late in the crash sequence.**

Analysis Update: Speed Data

- Accuracy: We typically apply +/-4% to speed data
- The NHTSA part 563 regulation specifies speed data MUST be accurate to within +/-1 kph (0.6 mph)
- This is for new cars, does not include tire wear which is -0/+2%

Analysis Update: Delta V Usage

VEHICLE WITH EDR

- Inline w/departure

  V1 = V3 - ΔVx

- Angled with Departure

  V1=V3cosΘ - ΔVx

Note no assumptions about whether collision was central or vehicles reached a common velocity

VEHICLE WITHOUT EDR

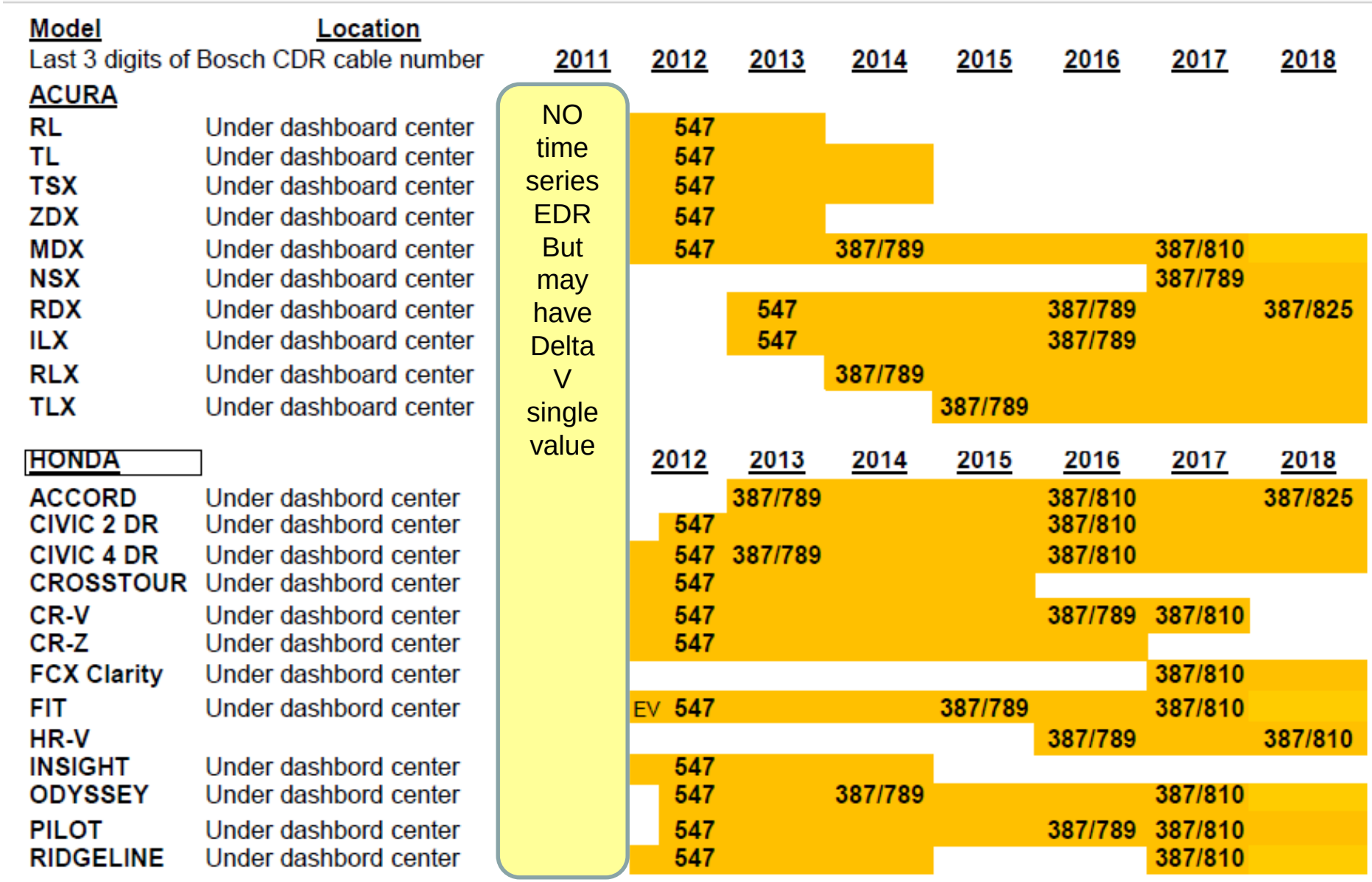
- Inversely Proportional DV at centers of mass $\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$
- Closing Speed Central (used inline)
- $\text{Closing Speed} = \left[\frac{1}{1+e}\right]\left[|\Delta V_1| + |\Delta V_2|\right]$ common velocity
- Closing Speed Offset (used inline)
- $\text{ClosingSpeed} = \left[\frac{1}{1+e}\right]\left[\frac{|\Delta V_1|}{\gamma_1} + \frac{|\Delta V_2|}{\gamma_2}\right]$ common velocity
- Intersection inline approximation (new)
- Triangular Velocity Vectors (relatively new)

Analysis Update: Delta V Data

- Ground Forces were typically ignored in reconstruction
- Especially for pedestrian and motorcycle hits into cars with EDR's, ground forces on the car become significant
- If car locks up brakes before hitting ped, Delta V of ped hit is overstated
- If MC t-bones car, Lat DV is understated

Details by Manufacturer

Honda/Acura CDR 17.5 Coverage 31%



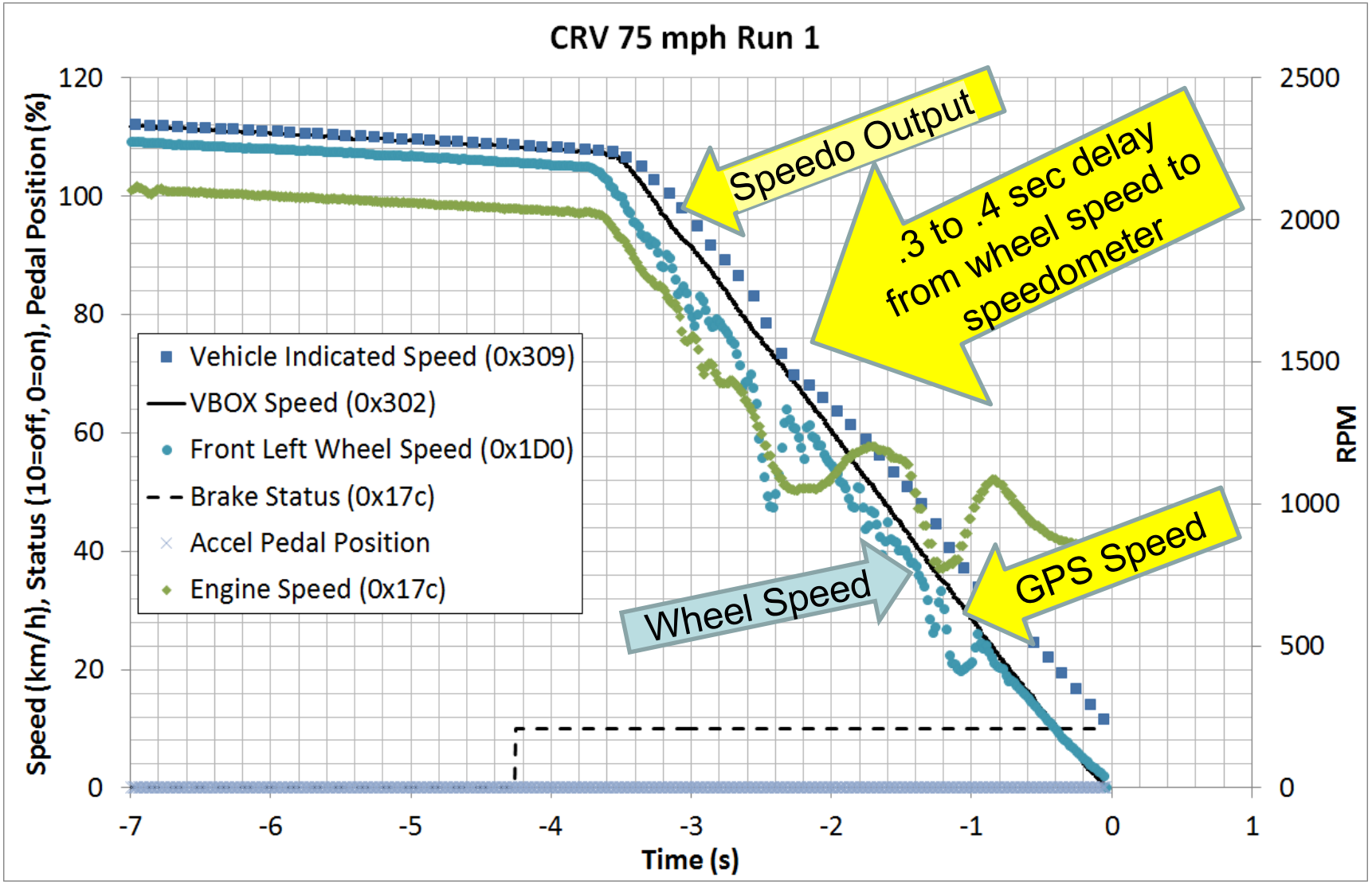
Honda EDR Evolution

- One set of DL's thru 2015
  - Roll rate polarity varies by supplier – polarity added under graph
  - Accel data instantaneous
- 2016-17 adds "Honda sensing" & Cruise
- 2018 – 3rd Gen to include "ig switch status"
  - Airbag deploys even if ign off until car stops
  - Markets outside US/Japan do not get precrash data

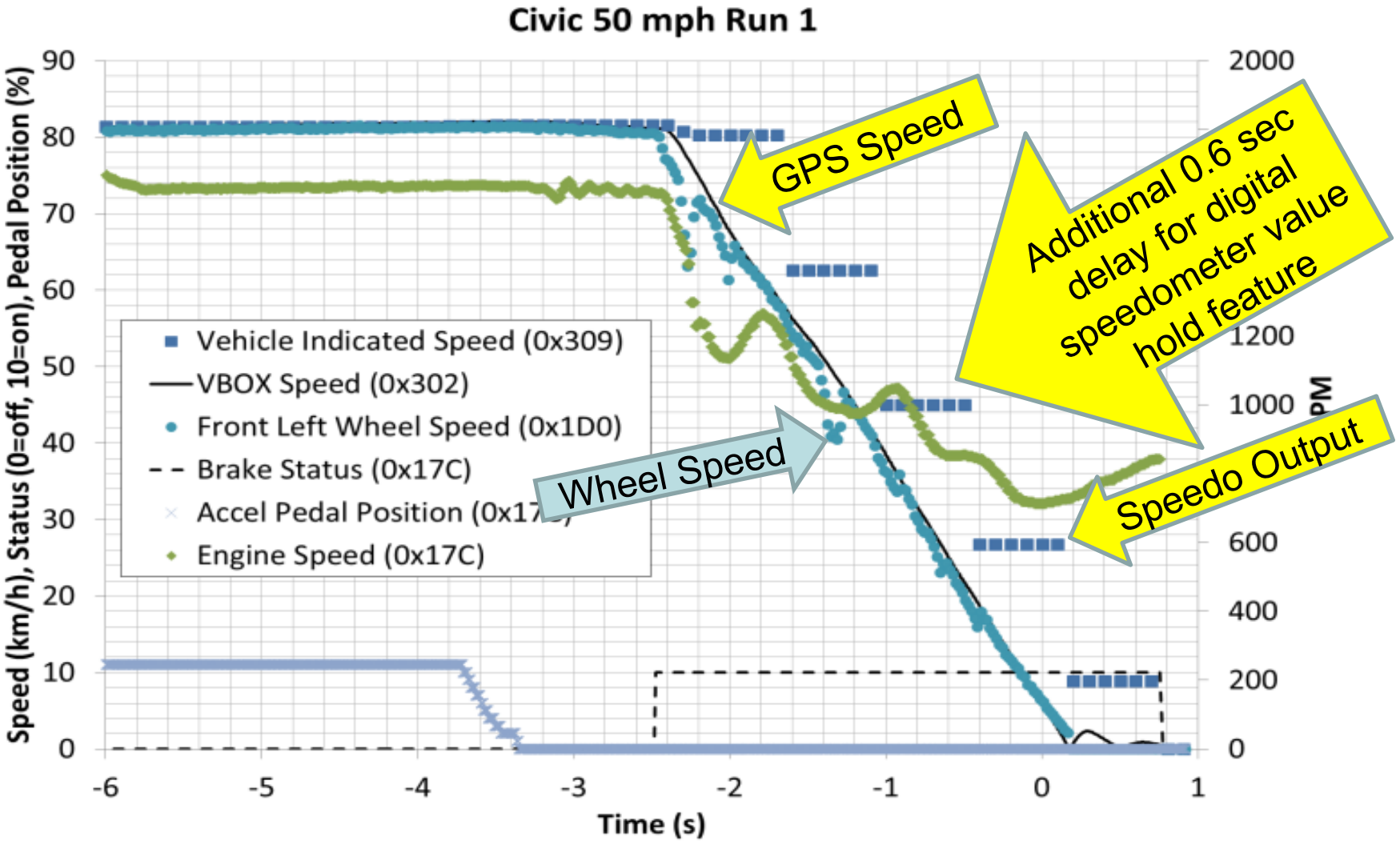
UNIQUE TO HONDA

- Most manufactures record the speed calculated in the powertrain control module as it is being sent **TO** the speedometer on the CAN bus.
- Honda records the speed being displayed **ON** the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
- The last data point before impact may be delayed more than it is from others

2012 CR-V Speed Reporting Delay



2012 Civic Speed Reporting Delay



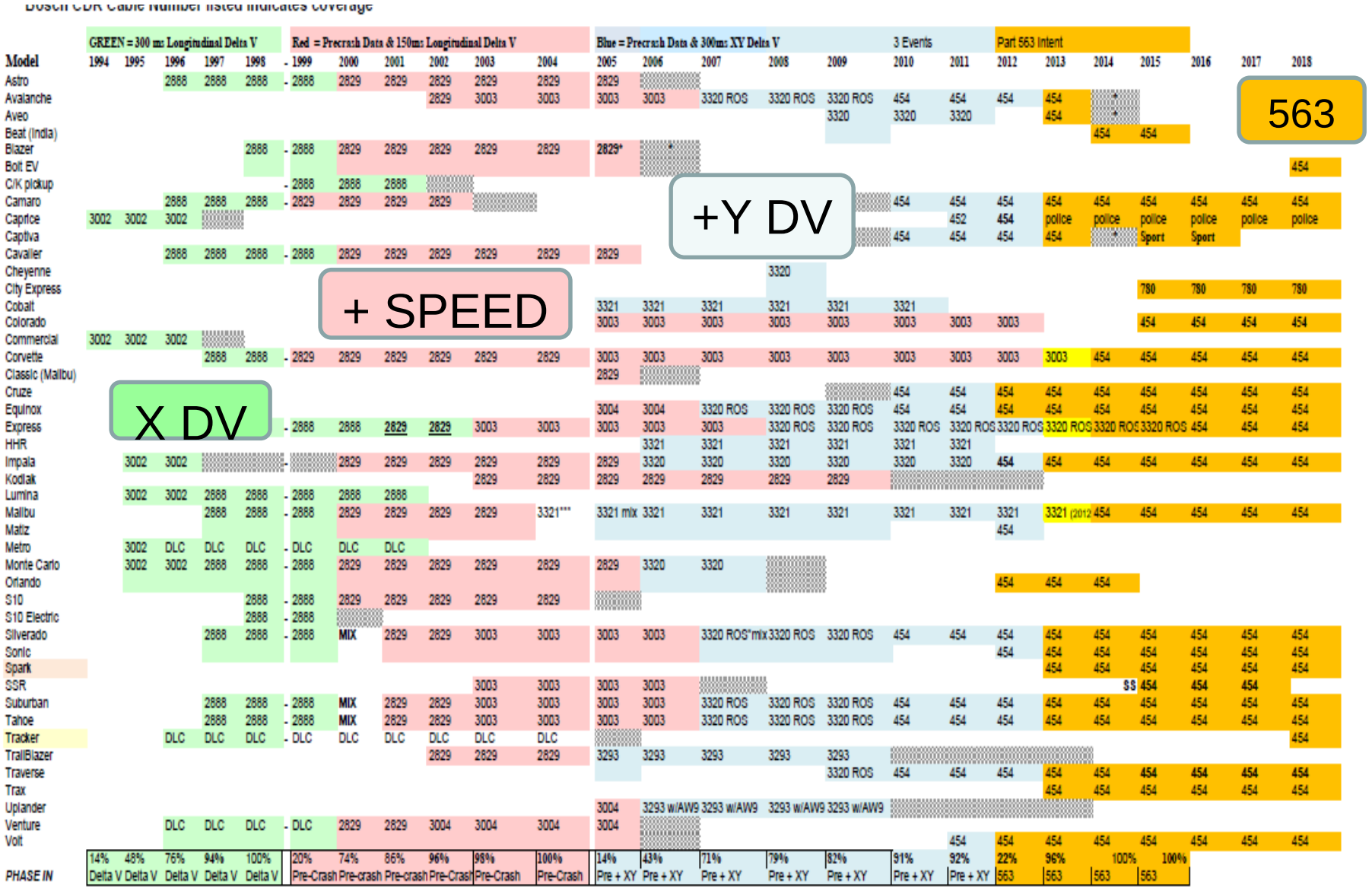
Honda Speed at Impact Calcs

- For ANALOG speedometer, assume last data point is SAMPLED 0.1 sec before AE, plus 0.3 sec filtering delay, adds up to 0.4 sec max delay from last data point to crash, use that in “MIN” column braking calcs.
- For DIGITAL speedometer, assume data is SAMPLED 0.1 sec before AE, plus 0.3 sec filtering delay, plus added 0.6 sec hold time. Add these to get 1.0 sec and use in “min” column braking calcs.

GM

- GM has so many models and brands, and for so many model years since 1994, the full timeline chart for just Chevrolet following is hard to read.
- A second chart follows showing only Chevrolet 2007 and later to make it more readable – the full GM chart is on my website www.ruthconsulting.com (as are the other manufacturer charts).

Chevrolet SDM Phase In **85%**



Chevrolet Recent Changes 85% 563

3 Events 1 animation | Part 563 Intent

Model	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Avalanche	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	*				
Aveo			3320	3320	3320		454	*				
Beat (India)								454	454			
Bolt EV												454
Camaro				454	454	454	454	454	454	454	454	454
Caprice					452	454	police	police	police	police	police	police
Captiva				454	454	454	454	*	Sport	Sport		
City Express									780	780	780	780
Cobalt	3321	3321	3321	3321								
Colorado	3003	3003	3003	3003	3003	3003			454	454	454	454
Corvette	3003	3003	3003	3003	3003	3003	3003	454	454	454	454	454
Cruze				454	454	454	454	454	454	454	454	454
Equinox	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454
Express	3003	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454
HHR	3321	3321	3321	3321	3321							
Impala	3320	3320	3320	3320	3320	454	454	454	454	454	454	454
Kodiak	2829	2829	2829									
Malibu	3321	3321	3321	3321	3321	3321	3321 (201	454	454	454	454	454
Matiz						454						
Orlando						454	454	454				
Silverado	3320 ROS*mix	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454
Sonic						454	454	454	454	454	454	454
Spark							454	454	454	454	454	454
SSR									$$ 454	454	454	
Suburban	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454
Tahoe	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454
Tracker												454
TrailBlazer	3293	3293	3293									
Traverse			3320 ROS	454	454	454	454	454	454	454	454	454
Trax							454	454	454	454	454	454

older family

Freeze Frame

No Precrash in 2013 Express/Savanna

GM Recent Changes

- 454 cable SDM's hold up to 3 events and do not require replacement after a D (until 3 D's). (2010+)
- Events numbered vs. labeled D/ND. Events may not be in order
- Does NOT erase ND's after 250 KC
- Overwrite other ND's after 250KC
- Acceleration data (see clipping!)
- Black plastic SDM housings

Other New GM News

- Some product liability data elements not relevant to reconstruction ***moved*** from CDR main report into **E**ngineering **T**ranslation **R**eports. Other elements do not display if not equipped (used to say “if equipped”)
- Euro Pedestrian protection coming to US
- Some 2011-16 Autoliv had single stage airbags (severity status NO for stage 2)
- Known “Features” noted in Data Lims
- Occupant classification only child vs adult (no midsize or large adult classification)

Important Recent Bug Fix

- **Some 2010-2012 GM (specifically all Autoliv SDM 10 & 11) interpreted *before* 17.0 (Aug 2016) misinterpreted some data**
- Brake status was reported ***backwards***
- Precrash was reported ***out of proper time order***
- **Fixed** in Software version 17.0
- As ALWAYS, be SURE to ***reinterpret data in the latest version of software!!!!!***

Specific models affected by brake and time order:

- **2010: SRX, Equinox, Terrain, Enclave, Captiva Sport, Traverse, Acadia, Outlook**
- **2011: SRX, Equinox, Terrain, Cruze, Volt, 9-4x, Enclave, Captiva Sport, Traverse, Acadia**
- **2012: Caprice, Impala, *Enclave, Captiva Sport, *Traverse, *Acadia**

- **Check closed cases????**

Important Recent Bug Fix 2

- In SDM 10 and 11 Autoliv, plus a few additional Autoliv models, the Roll Rate polarity if interpreted before 17.0 may be the opposite of SAE polarity
- **Affected models include Autoliv SDM-10/10i, SDM-10P, SDM-11, SDM-11e, SDM-11P and SDM-30.**

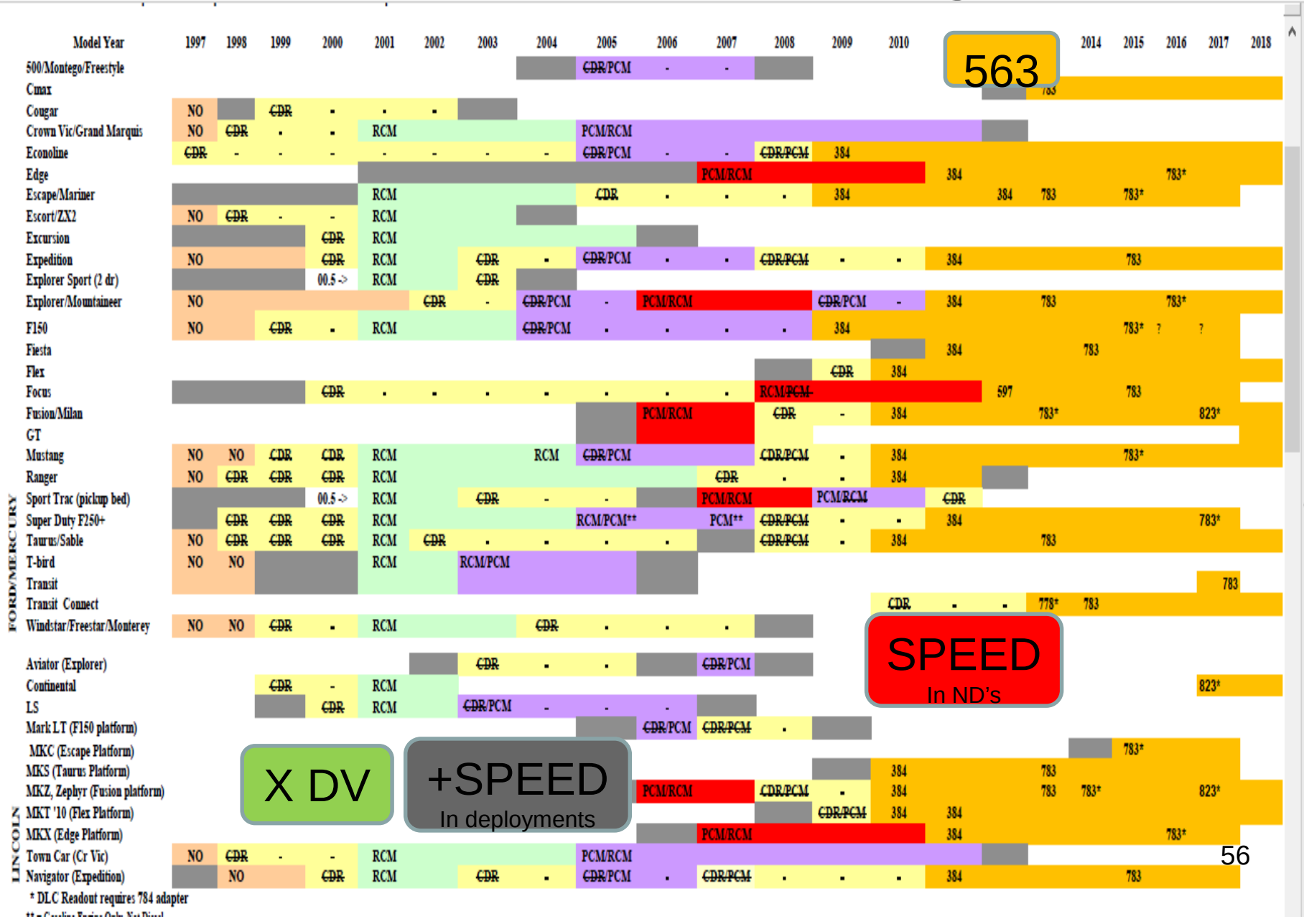
Other GM News

- One module family Conti SDM-30 comes out misinterpreted – data is scrambled
- 454 cable modules list accel limits
- Must send to Conti for manual readout and interp. (ask instructors for how)
- Foreign coverage – China since 99 (but only Gov't/NPS can use CDR), Australia/Korea/Thailand since 07
- South America not covered or spoofable

GM Ignition Switch Recall

- The GM Ignition Switch Recall where ignition power may cut out and cause loss of power steering continues to be unfairly blamed as the cause of crashes.
- The older red EDR group SDM does not write an event or deploy airbags after the key is off
- The new blue EDR group SDM may write an event for 3.25-4.5 seconds after key off, but airbag will not deploy if key was off – events will be recorded as ND’s. (Epsilon only Conti)
- Consult Haight article in Collision for details

Ford CDR 17.5 Coverage **59%**



Ford EDR evolution

- Autoliv RC5 (06)-RC6(09)-RC6X(11)-RC7(13)
- Bosch AB9(05)-AB10(12)-AB11(17) AB12(17)
- Chrysler had stability control yaw rate and steering earlier, but Ford upped the game in 2009 with RC6 longitudinal and lateral accel data at 0.1 second intervals pre-crash
- Fords have rich data, but are skimpy on data limitations until the 2013 Taurus Autoliv RC7.
- Critical info only found in training materials

Polarity on Stability Control 09+

- The steering and yaw rate are opposite the normal right hand rule as documented in SAE 2014-01-0504.
- The paper reported lateral acceleration may be OK, but subsequent empirical data showed lateral acceleration is **also** backwards. In a hard right swerve, all 3 are negative.
- New RC7 and AB12 DL's show the polarity as stated above, all 3 backwards

09+ Ford Stability Control Data

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)

Times (sec)	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)	Stability Control Yaw Rate (deg/sec)
- 1.4	0.086	-0.045	0.12
- 1.3		-0.031	1.5
- 1.2		-0.647	-1.5
- 1.1		-0.75	-2.5
- 1.0		-0.709	-1.5
- 0.9		-0.763	-1.5
- 0.8		-0.826	-1.5
- 0.7		-0.792	-0.75
- 0.6		-0.895	1.25
- 0.5	0.014	-0.83	1.87
- 0.4	0.03	-0.835	0.75
- 0.3	0.022	-0.607	0.37
- 0.2	0.107	-0.653	0.75
- 0.1	0.05	-0.823	-3.37
0.0	-0.004	-0.811	-3.75

Integrate lat accel - get velocity – integrate again – get displacement

Reaction

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

Ford Occupant Size Classification

not in data limitations

Pre-Crash Data -1 sec (First Record)

Ignition cycle, crash	359
Frontal air bag warning lamp, on/off	OFF
Occupant size classification, front passenger (Child size Yes/No [Hex value])	No [$08]
Safety belt status, driver	Driver Buckled

- The Hex Value provides additional information that just child yes/no
- The Decoder is different for Ford's 2 main suppliers, Autoliv and Bosch
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

02011_RCM-AB10P_r001

02007_RCM-RC6_r002

New Second Decoder for Bosch

Autoliv	**Bosch**	
$01	$00	Empty
$02	$01	Child (clearly 55 lbs or less)
$04	$05	Indeterminate (gray zone somewhere between 55 to 110 lbs)
$08	$02	Small Adult (not a child and always registers for 110 lb 5th female adult)
$10	$03	Medium adult (not a 5th female and always registers for what is probably a 50th male adult of 170 lbs)
$20	$04	Large Adult (next bigger threshold than above, 95th male , 223 lbs)

Autoliv RC6 Anomalies (09-10)

- There are timing delays in reporting the 0.5 sec interval precrash data in the early Ford Autoliv RC6 modules (09 F150-Escape-Econoline and many 2010). Identify by lag between stability control longitudinal (braking) g's and brake on time and speed lost. Use Stability Control System long. accel for timing of accel release and brake on.
- Fixed by "RC6X", identifiable only by opening .cdrx in notepad, observe "RC6_2011" in header (and by agreement between stability control and 0.5 sec interval precrash data).

AB10 Anomaly

- The “Bosch AB10” ACM found in 2012+ Focus and 2015 Escape and other vehicles **produced before Nov. 2015**, which have a **second event**, incorrectly label pre-crash **time** to impact in the second event.
- Time between events is accurate – use it
- Second event precrash (both speed and stability control sections) include extra points at irregular intervals labeled as if they occurred at regular intervals.

AB10 Anomaly cont'd

- If you can see the first event in the second event's data, it will lead you to think the time between events is longer than it really is.
- The data in the second event did occur, just not at the time it said it did.
- The second event last precrash speed is still within 0.5 seconds of the second event.
- There is no software fix for EDR's produced before Nov. 2015. Fix implemented 11/2015.
- No mention in DL's yet.

Evolution: Getting Even Better

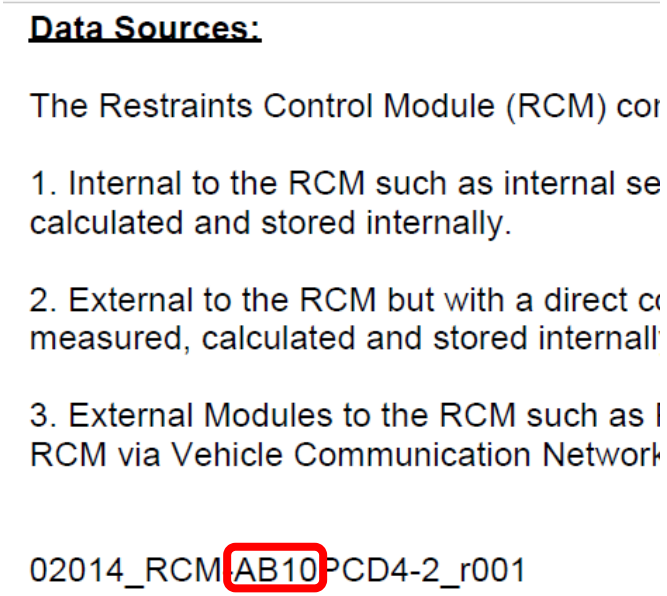
- RC-7 module that has larger RAM memory and buffers speed data every 0.1 seconds.
- When an event is set, it goes back and takes the most recent value and labels it “0.0”. Values at 0.5 sec intervals are picked from buffer relative to 0.0.
- Data point are within 0.1 seconds of what they are labeled
- Good set of data limitations
- Some driver assistance data elements

Bosch AB12 ACM - 2017 MKZ

- Precrash data ***every 0.2 seconds*** instead of every 0.5 seconds
- Stability control data steering/yaw are on a different page than lat/long accel are (need to cut/paste columns to use current analysis template)
- Report has page for postcrash data 0 to +5 at 0.25 sec intervals , but only populated with “Event in Progress” at this time
- Some Driver Assistance data elements

Identifying Ford EDR families

- At the end of the data limitations section, just below the last paragraph, on the left, Bosch puts what version of data limitations is being printed. For Ford, most data limitations by family.
- No difference between RC6 and RC6X
- Works on GM too



For Ford EDR **not supported** by Bosch CDR See the document on my website "Unsupported Fords"

Many requests can go directly to Ford's suppliers.

Other requests for assistance to Ford legal:

Email to EDRFORD@FORD.COM

(this is an address for a Bowman and Brooke outside counsel that handles these requests for Ford)

Contact Rick Ruth for additional information prior to making request

As of this writing, Autoliv charges $800 to all, Bosch and Conti are no charge to law enforcement but $1000 to private parties.

Recent 05 Freestar RCM sent to Takata was returned with Hex Data only and no guide to interpretation (no charge, no value!)

Available on my website:

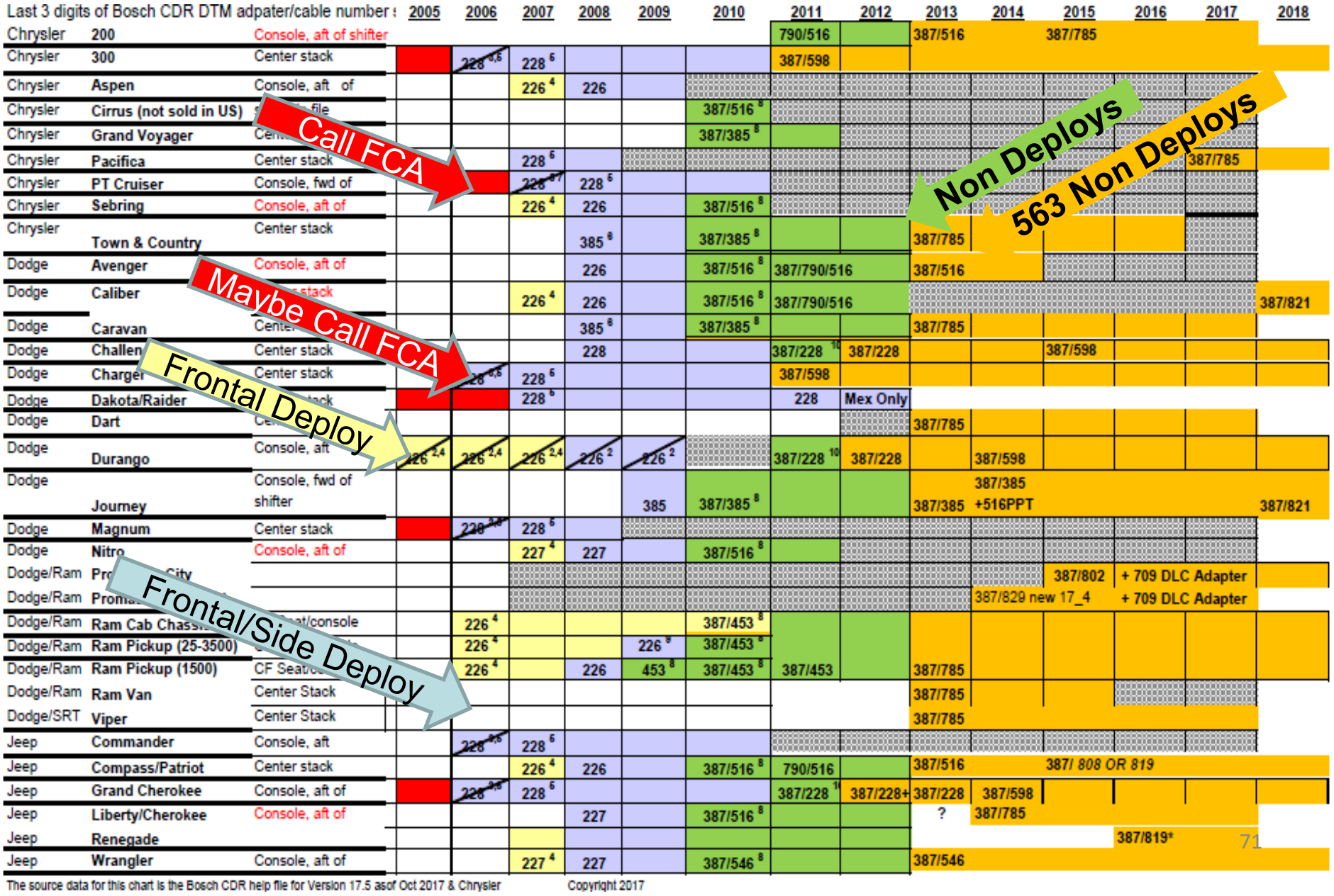
Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 4-12-2017, after the original presentation at the Jan 2016 CDR Summit)



FCA 17.5 CDR Coverage 51%



Last 3 digits of Bosch CDR DTM adpater/cable number			2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Chrysler	200	Console, aft of shifter							790/516		387/516		387/785			
Chrysler	300	Center stack		228 [5,6]	228 [5]				387/598							
Chrysler	Aspen	Console, aft of			226 [4]	226										
Chrysler	Cirrus (not sold in US)	file						387/516 [8]								
Chrysler	Grand Voyager	Cen						387/385 [8]								
Chrysler	Pacifica	Center stack			228 [5]										387/785	
Chrysler	PT Cruiser	Console, fwd of			226 [4,7]	228 [5]										
Chrysler	Sebring	Console, aft of			226 [4]	226		387/516 [8]								
Chrysler	Town & Country	Center stack				385 [8]		387/385 [8]			387/785					
Dodge	Avenger	Console, aft of				226		387/516 [8]	387/790/516		387/516					
Dodge	Caliber	stack			226 [4]	226		387/516 [8]	387/790/516							387/821
Dodge	Caravan	Center				385 [8]		387/385 [8]			387/785					
Dodge	Challen	Center stack				228			387/228 [10]	387/228			387/598			
Dodge	Charger	Center stack		228 [5,6]	228 [5]				387/598							
Dodge	Dakota/Raider	tack			228 [5]				228	Mex Only						
Dodge	Dart	Ce									387/785					
Dodge	Durango	Console, aft	226 [2,4]	226 [2,4]	226 [2,4]	226 [2]	226 [2]		387/228 [10]	387/228		387/598				
Dodge	Journey	Console, fwd of shifter					385	387/385 [8]			387/385	387/385 +516PPT				387/821
Dodge	Magnum	Center stack		228 [5,6]	228 [5]											
Dodge	Nitro	Console, aft of			227 [4]	227		387/516 [8]								
Dodge/Ram	Pro City												387/802	+ 709 DLC Adapter		
Dodge/Ram	Promaster											387/829 new 17_4		+ 709 DLC Adapter		
Dodge/Ram	Ram Cab Chass	at/console		226 [4]				387/453 [8]								
Dodge/Ram	Ram Pickup (25-3500)			226 [4]			226 [8]	387/453 [8]								
Dodge/Ram	Ram Pickup (1500)	CF Seat/		226 [4]		226	453 [8]	387/453 [8]	387/453		387/785					
Dodge/Ram	Ram Van	Center Stack									387/785					
Dodge/SRT	Viper	Center Stack									387/785					
Jeep	Commander	Console, aft		228 [5,6]	228 [5]											
Jeep	Compass/Patriot	Center stack			226 [4]	226		387/516 [8]	790/516		387/516		387/ *808 OR 819*			
Jeep	Grand Cherokee	Console, aft of		228 [5,6]	228 [5]				387/228 [10]	387/228+	387/228	387/598				
Jeep	Liberty/Cherokee	Console, aft of				227		387/516 [8]			?	387/785				
Jeep	Renegade													387/819*		
Jeep	Wrangler	Console, aft of			227 [4]	227		387/546 [8]			387/546					

The source data for this chart is the Bosch CDR help file for Version 17.5 asof Oct 2017 & Chrysler

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Chrysler Data

- Chrysler has more frequent precrash data with 50 data points at 0.1 sec intervals in most models (0.2 sec in old minivans)
- No events in ND's until 2009/10, Accel data instead of Delta V in early models
- Key cycle info phased in starting 2010, X and Y Delta V starting 2011, seat belt use and deploy times also phased in, some 2011+ also have odometer readings.
- **Report format changed recently to make some data at 0.25 sec intervals to make room for other data to be recorded**

Output New Look – more on Pg1

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full	Service Brake	Engine RPM	ABS Activity	Stability Control	Steering Input (deg)
-5.0	Complete	96 [155]	0.00	Off	3,878	No	On	1
-4.9	Complete	96 [154]	0.00	Off	3,835	No	On	-1
-4.8	Complete	96 [154]	0.00	Off	3,726	No	On	-5
-4.7	Complete	96 [154]	0.00	Off	3,621	No	On	-10
-4.6	Complete	96 [154]	0.00	Off	3,489	No	On	-18
-4.5	Complete	95 [154]	0.00	Off	3,344	No	On	-21
-4.4	Complete	95 [153]	0.00	Off	3,210	No	On	-20
-4.3	Complete	95 [153]	0.00	Off	3,109	No	On	-19
-4.2	Complete	95 [153]	0.00	Off	3,068	No	On	-19
-4.1	Complete	95 [152]	0.00	Off	3,108	No	On	-18
-4.0	Complete	95 [152]	0.00	Off	3,079	No	On	-18

Wheel RPM replaced by speed

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	ABS MIL	Yaw Rate (deg/sec)	Wheel Speed LF (km/h)	Wheel Speed RF (km/h)	Wheel Speed LR (km/h)	Wheel Speed RR (km/h)
-5.0	Off	0.40	154.37	154.52	154.55	154.52
-4.9	Off	-0.48	154.28	154.16	154.25	154.25
-4.8	Off	-1.76	154.40	154.09	153.88	153.77
-4.7	Off	-2.80	154.28	153.95	153.88	153.67
-4.6	Off	-4.24	154.01	153.64	153.85	153.55
-4.5	Off	-6.80	153.98	153.16	153.80	153.19
-4.4	Off	-7.12	153.80	152.88	153.70	152.80
-4.3	Off	-5.84	153.49	152.67	153.37	152.62
-4.2	Off	-5.12	153.13	152.19	153.25	152.38
-4.1	Off	-4.64	152.80	152.04	152.91	151.91
-4.0	Off	-4.64	152.49	151.73	152.46	151.65
-3.9	Off	-4.00	152.13	151.55	152.13	151.31
-3.8	Off	-3.68	151.83	151.25	151.95	151.13

Current Gear & More @ 0.25 sec

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

Time Stamp (sec)	Pre-Crash Recorder Status	Raw Manifold Pressure (kPa)	PCM MIL	ETC Lamp	Current Gear	Reverse Gear Status
-5.00	Complete	20.00	Off	Off	D5	Not_Reverse
-4.75	Complete	20.00	Off	Off	D5	Not_Reverse
-4.50	Complete	21.60	Off	Off	D5	Not_Reverse
-4.25	Complete	23.20	Off	Off	D5	Not_Reverse
-4.00	Complete	23.20	Off	Off	D6	Not_Reverse
-3.75	Complete	23.20	Off	Off	D6	Not_Reverse
-3.50	Complete	23.20	Off	Off	D6	Not_Reverse
-3.25	Complete	23.20	Off	Off	D6	Not_Reverse
-3.00	Complete	22.40	Off	Off	D6	Not_Reverse
-2.75	Complete	21.60	Off	Off	D6	Not_Reverse
-2.50	Complete	21.60	Off	Off	D7	Not_Reverse
-2.25	Complete	21.60	Off	Off	D7	Not_Reverse
-2.00	Complete	21.60	Off	Off	D7	Not_Reverse
-1.75	Complete	21.60	Off	Off	D7	Not_Reverse

x 10.63 in

Tire pressure @ 0.25 sec intervals

NEW

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Tire Pressure Position	Tire Pressure Status	Tire Pressure Value (PSI)	Cruise Control Lamp Status	Cruise Control Engaged Status	ACC On/Off	ACC Set Speed (MPH [km/h])
-5.00	LHF	Normal	34	Off	Not_Engaged	Off	0 [0.0]
-4.75	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.50	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.25	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.00	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-3.75	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.50	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.25	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.00	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.75	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.50	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]

Integrating Yaw Rate

From prior page

- 3 deg/sec * 0.1 sec =	-0.3 degrees
-11 deg/sec * 0.1 sec =	-1.1 degrees
-18 deg/sec * 0.1 sec =	-1.8 degrees
-15 deg/sec * 0.1 sec =	-1.5 degrees
- 9 deg/sec * 0.1 sec =	-0.9 degrees
- 4 deg/sec * 0.1 sec =	-0.4 degrees
- 1 deg/sec * 0.1 sec =	-0.1 degrees
Total Swerve Right	6.1 degrees

Note this is how far nose of car is pointing right – *CG* motion may *lag*

Pedestrian Protection

- Starting 2014, In Europe only, FCA has added a pedestrian protection feature to the Dodge Journey, Fiat Freemont, and Lancia Grand Voyager to deploy a feature that pops the hood up slightly to make the angle better and less injurious to pedestrians. Toyota and many others have followed.
- It is a separate module with separate data and must be imaged separately thru the DLC or via cable 516.
- Widespread in Europe, coming to US soon

Other FCA News

- Revised reporting of deploy times (was stage 1 to stage 2 vs AE to stage 2).
- 17.2 fixed WOT issue reporting as 77%
- 10-12 Journey + Minivan – all ND's display as "interrupted" recording
- 2013 TRW – key cycle counters may not match perfectly (different sources)
- Recording threshold intent 5mph DV, but some models may report a 4.9 DV.
- Most "features" are noted in data limitations. i.e recordings under 5mph DV

Chrysler Data Limitations

- **First** set for all 2005-2012 Models
- **Second** set for all 2013-current Models
- Read thoroughly, often describes the most common data first then lists exceptions after the general rule.
- Many limitations are for OTHER MODELS than the one you are investigating.
- Due to some non-US production, there are some unusual models

New Chrysler Info

- Frontal sensors are 100G at 45 degrees (100*cos 45° = 71g straight ahead).
- Replaced clipping flag with “Sensor Design Range Exceeded” line on graph
- Buckle switch NOT normally used to determine if pre-tensioner fires (Exceptions on some non US built vehicles)
- Fiat Spider = Mazda Miata, event threshold less than 8kph
- Driver assistance data elements coming in

Toyota EDR Families

First Generation

- 00/02 with frontal crash only recording
- 00/02 with front and side/roll if equipped

Second Generation

- 04/06 EDR

Gen 1&2: Precrash data is not tied to a module family.

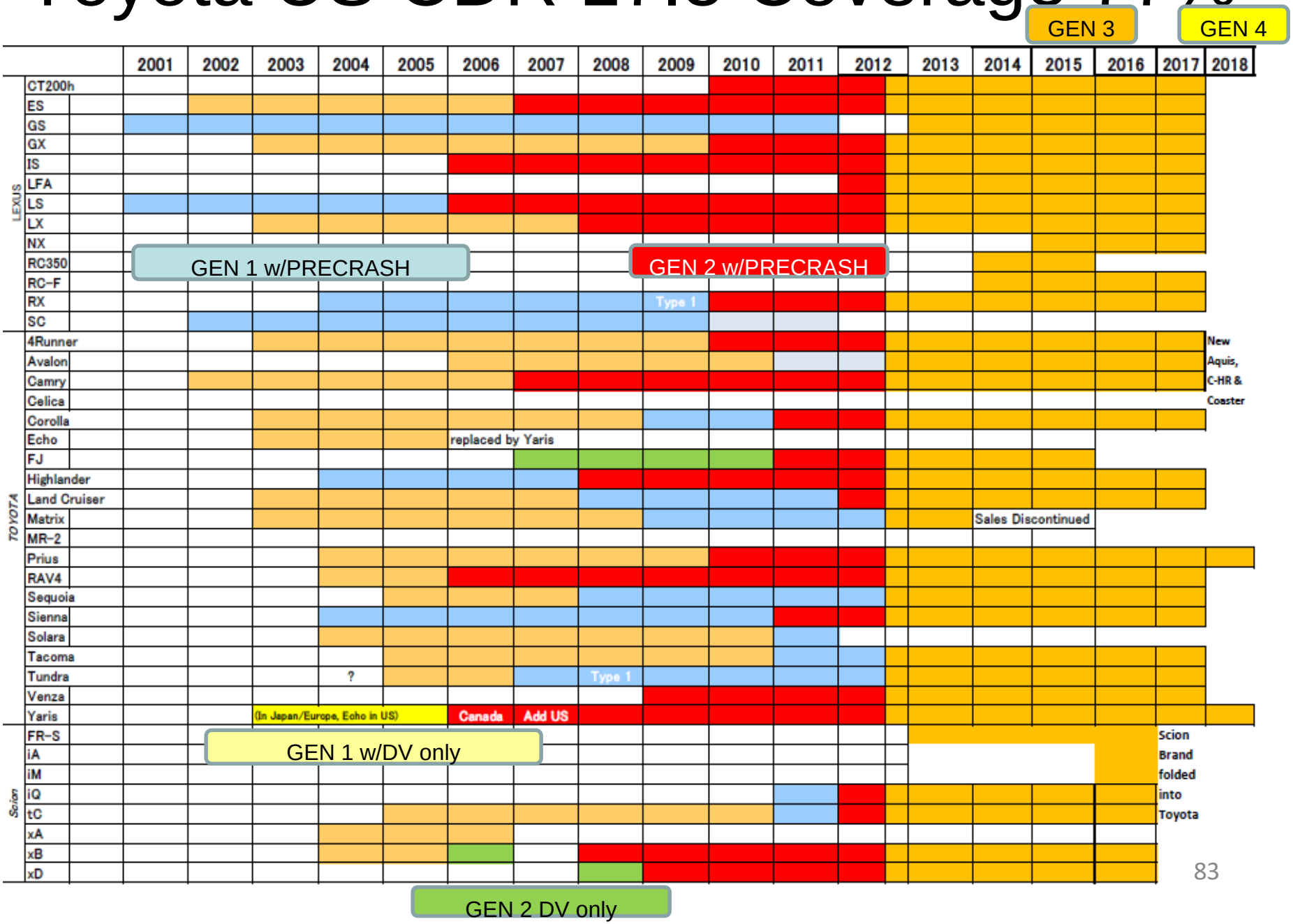
Third Generation (in US)

- 12 EDR = 563-intent, no RCM Lat DV in frontal events.
- 13 EDR, 563 intent, RCM Lat accel in frontals
- 15 EDR, adds side impact door pressure sensors

Fourth Generation

- 17 EDR (2018 models)

Toyota US CDR 17.5 Coverage 77%



Toyota EDR Evolution

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
3 front 1 side 1 roll	2 front 2 side 2 roll	2 front 2 side 2 roll	4 front 4 side 6 precrash
Record on 2g wakeup	Record on 2g wakeup	Record on 5mph XDV, side 3.5g wake	Record on 5 X or Y DV
150ms XDV 5PC@1 + TRG	200ms XDV 5PC@1+TRG	250ms XDV 10PC@0.5 + TRG	250ms XY 10PC@0.5+TRG

Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
70ms YDV@AC M+sats	70ms YDV@AC M+sats	12 YDV 70ms sats only 13 200ms ACM	250ms YDV
REFRESH 500 MS SPEED	REFRESH 500 MS SPEED	REFRESH 24 MS SPEED	REFRESH 24 MS SPEED
78 MPH TOP, 2kph res	75 MPH TOP. 2kph res	122 MPH TOP 1 kph res	122 MPH TOP, 1 kph res
Accel Pedal Volts	Accel Pedal Volts	Accel Pedal % & Throttle %	Accel Pedal % + Throttle

Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
RPM to 400	RPM to 400	RPM to 100	RPM to 100
No Key Cycles	No key cycles	Key Cycles	Key cycles
Time Between Events < 5 sec	Time Between Events <16 sec	Time Between Events <16 sec	Time Between Events <16 sec
		Brk Oil Pres, Stab. Cont. Long Accel	Brk Pres, Long Acc
		Strg & Yaw	Strg & Yaw

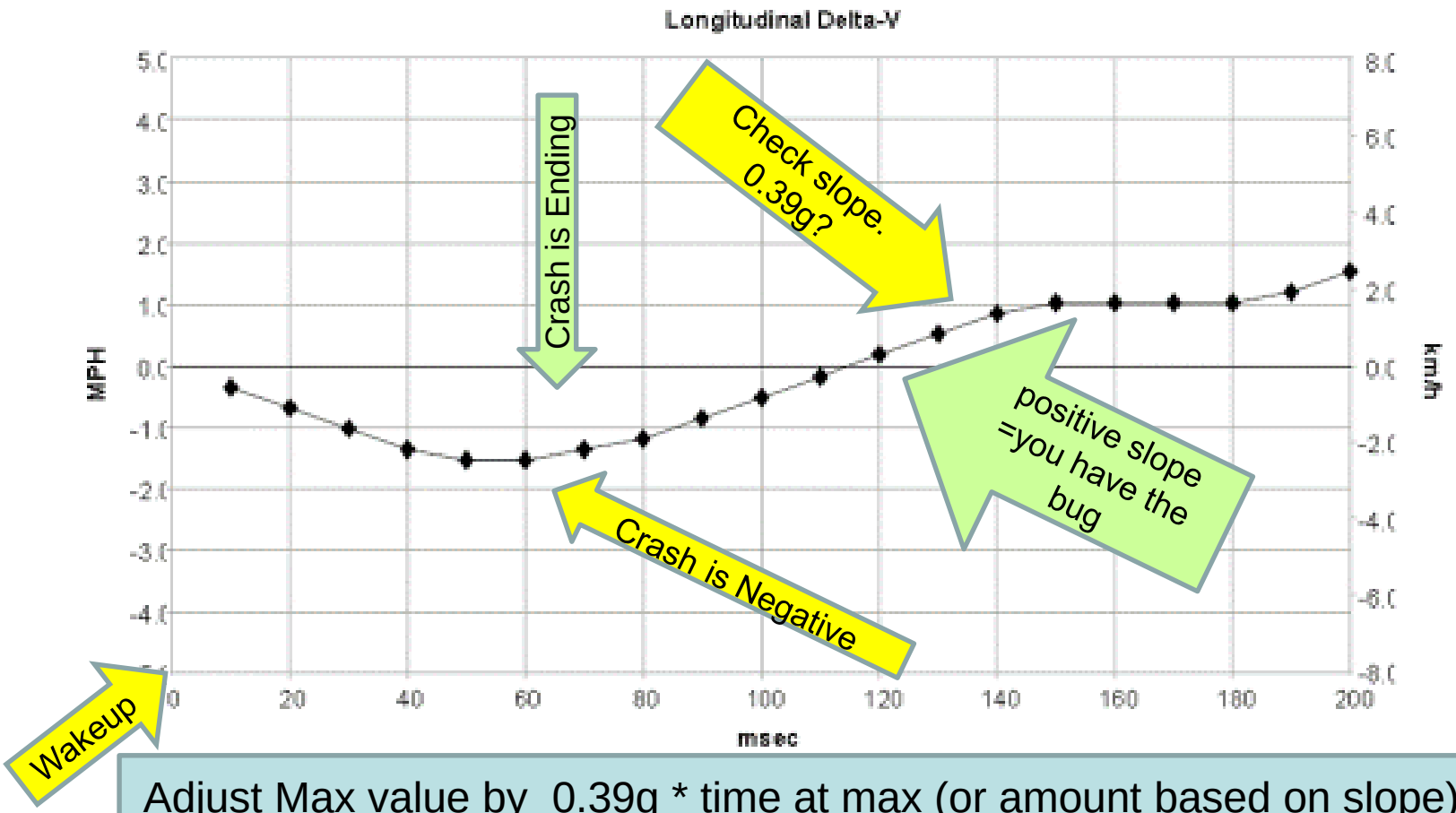
Known Bug in Gen 1&2 Delta V

- Empirical research indicates in all Gen 1's tested to date and SOME Gen 2 models
- ***Under*** reports *frontals* slightly
- ***Over*** reports *rears* slightly
- Sources in Toyota indicate the zero-offset applied when converting analog to digital accelerometer output may be off by 1 count, creating an error of 0.39G.
- 0.39g*150msec*32.2/1.466 = 1.3 mph
- Empirical data suggests 2 counts on some

How to Identify you have the bug

Longitudinal Crash Pulse (Most Recent Event, TRG 3 - table 1 of 2)

Recording Status, Time Series Data	Complete
Max Longitudinal Delta-V (MPH [km/h])	1.5 [2.5]



Adjust Max value by 0.39g * time at max (or amount based on slope)

No positive slope = no adjustment needed

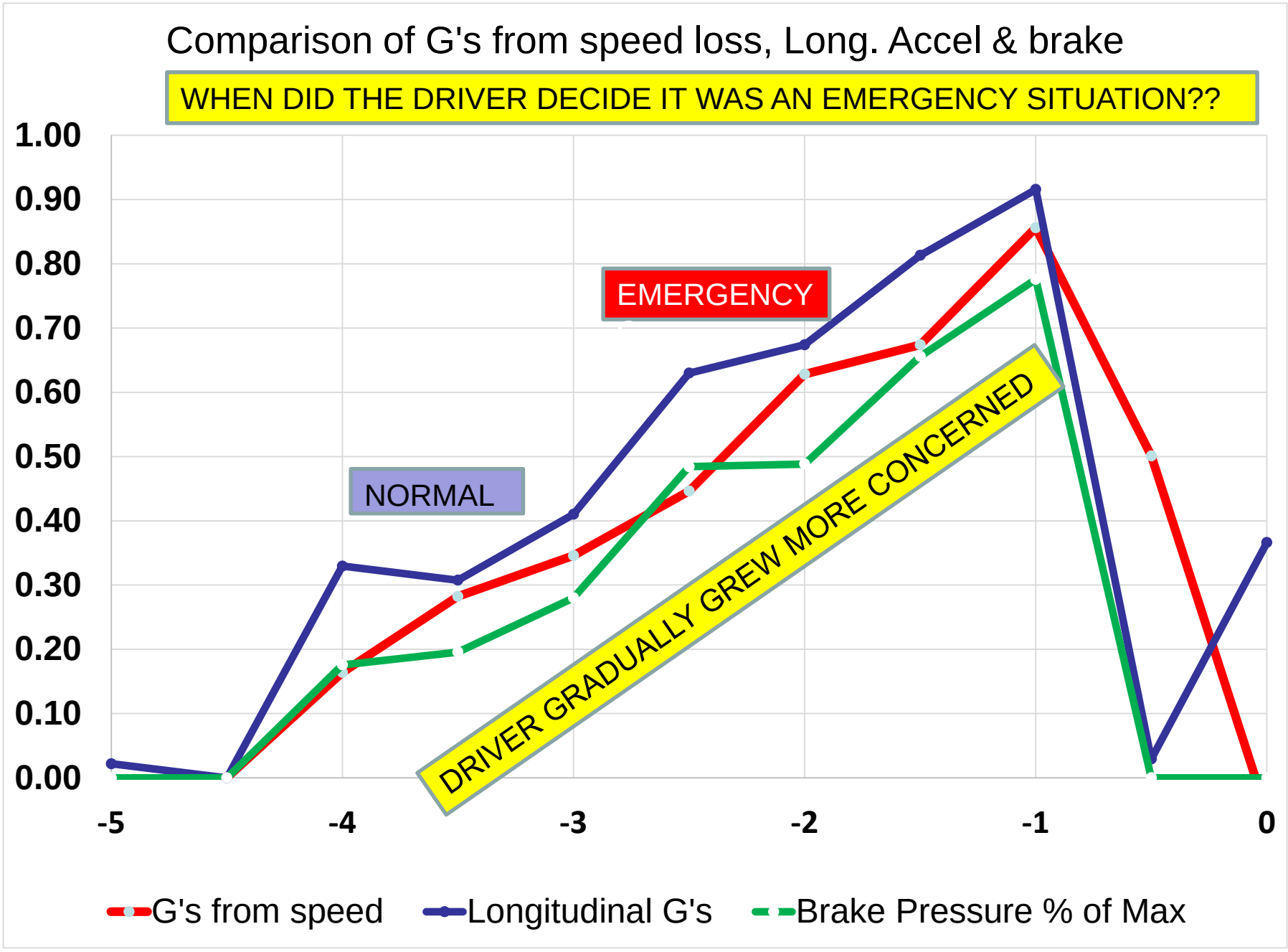
Toyota 12 EDR Drawback

- Lateral acceleration is only recorded in a side algorithm wakeup.
- When it is recorded, it is only for 68ms
- It is only recorded for the side satellites, not for the central control module
- If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location
- Fixed in “13EDR” – get 250ms of Y accel

3 ways to get braking rate

- Take consecutive speed data points, subtract them to get speed change vs time, divide by accel due to gravity to get how many G's. For -1.5 sec:

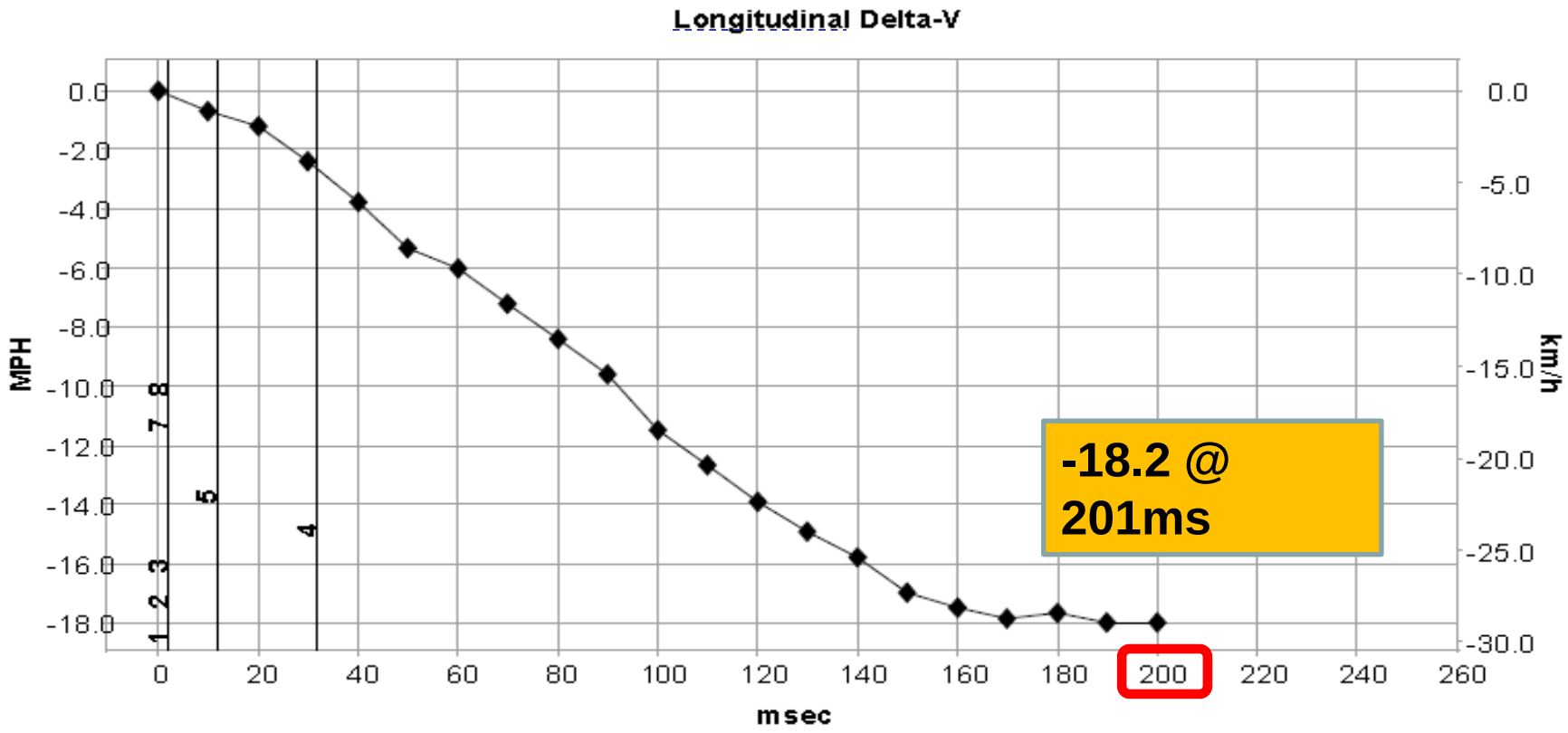
  (29.8-22.4) = 7.4 mph*1.466/32.2/0.5 sec = **0.67G**
- Compare this to longitudinal acceleration in m/s^2, divide by accel due to gravity of 9.8 m/sec^2 to get G's Example: -7.968m/sec^2 / **9.8** = **0.82G**
- Compare this to % max brake pressure in Mpa **7.87**/12 = 66%
- See next slide for graph **comparing** 3 methods


Comparison of G's from speed loss, Long. Accel & brake
WHEN DID THE DRIVER DECIDE IT WAS AN EMERGENCY SITUATION??
1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
-5
-4
-3
-2
-1
0
EMERGENCY
NORMAL
DRIVER GRADUALLY GREW MORE CONCERNED
G's from speed
Longitudinal G's
Brake Pressure % of Max

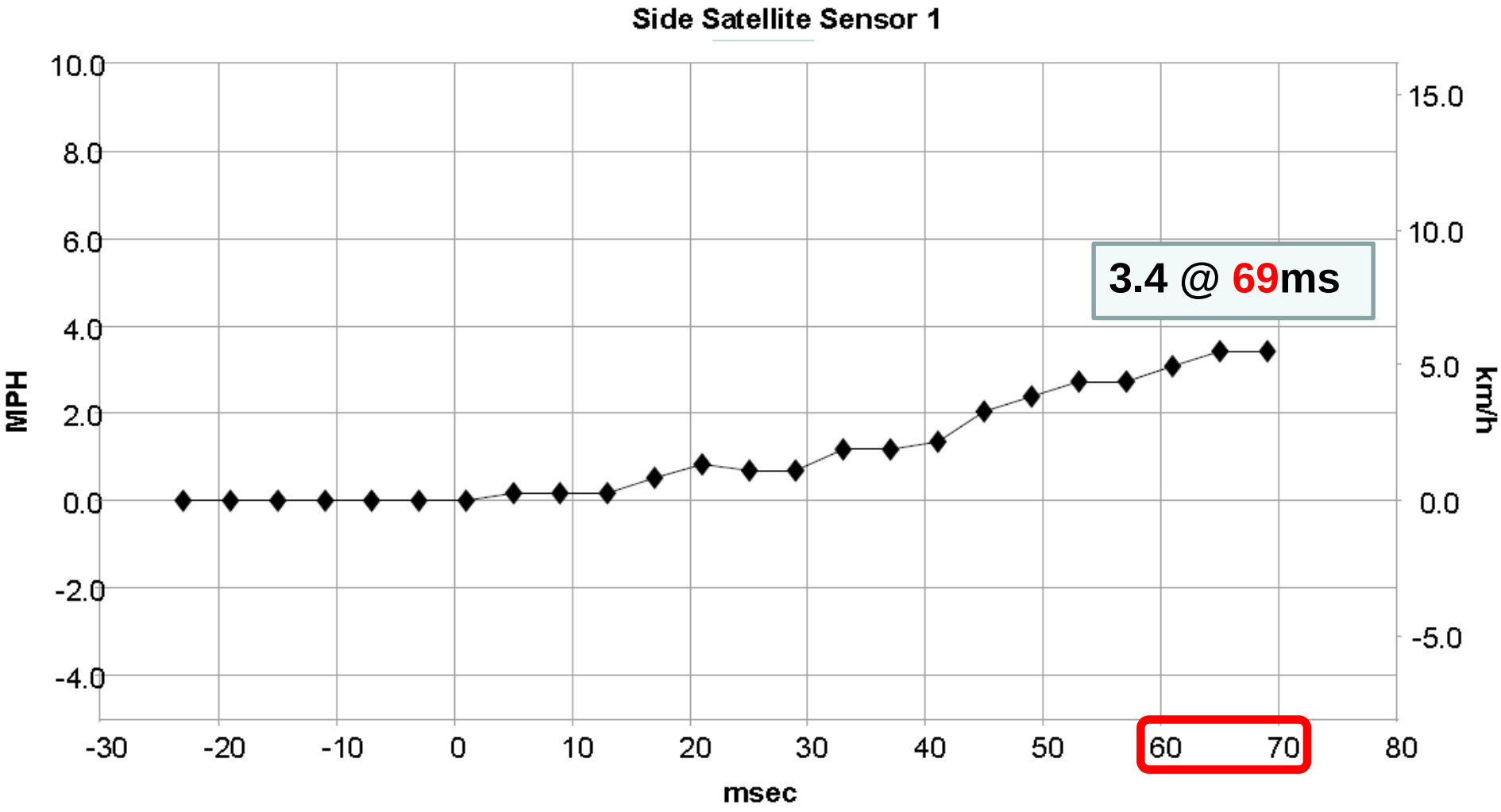
Delta V analysis – What's the PDOF? The Total?? 12 EDR

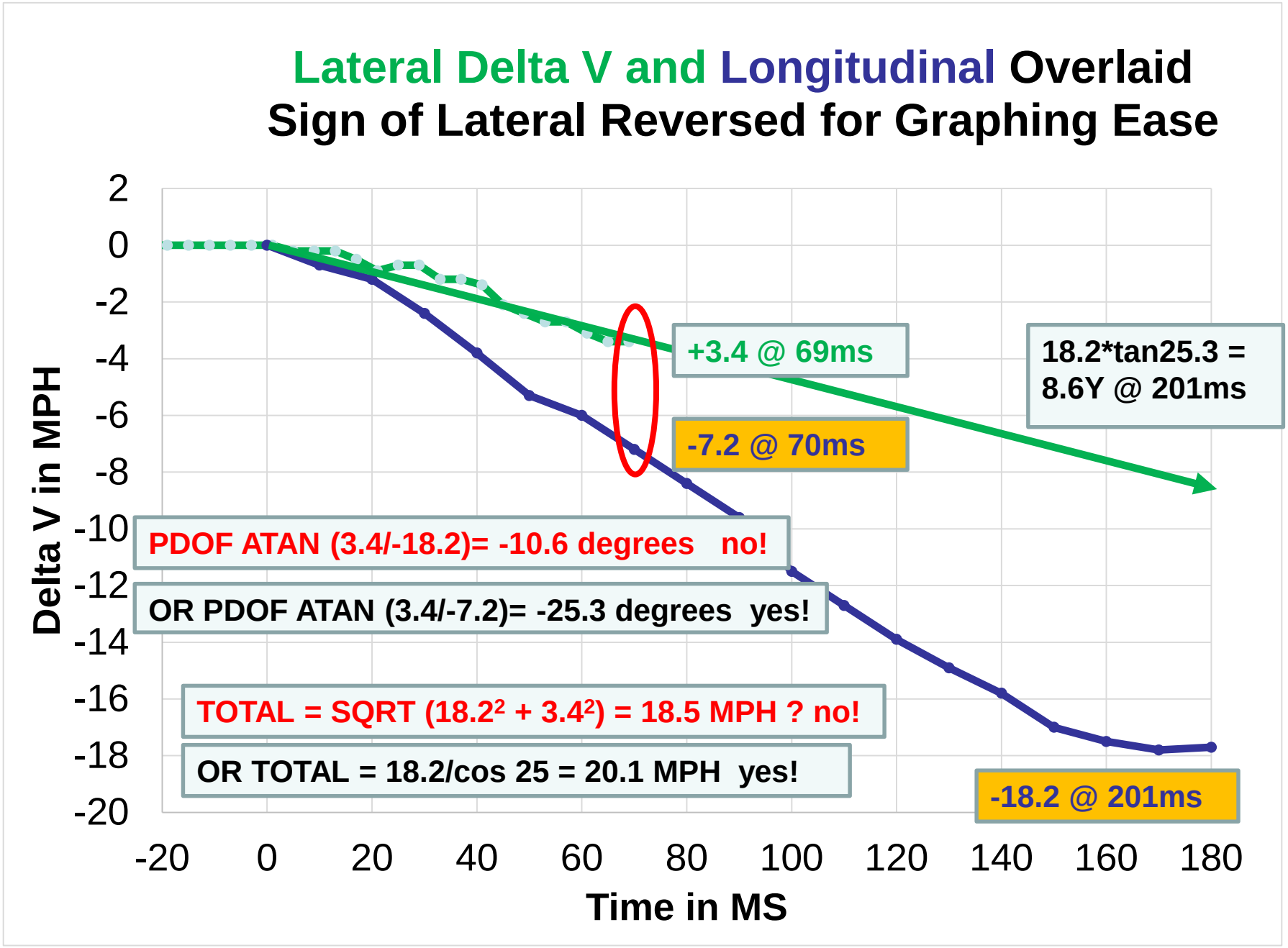
Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	2.0
Length of Delta-V (msec)	200
Max. Longitudinal Delta-V (MPH [km/h])	-18.2 [-29.2]
Time, Maximum Delta-V, Longitudinal (msec)	201.0
Power Supply Status at Max. Delta-V	ON



12 EDR Lateral Delta V -




Lateral Delta V and Longitudinal Overlaid
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees yes!
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

New “Vehicle History Utility”

- New feature in 2013/4 Rav 4 and 2014 Highlander, IS250/250, 2015 Prius, NX200t, NX300h, RC350, RCF
- Triggers EDR-style recording (5-8 secs pre and post) if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion (26 triggers)
- Accessible only with Toyota Techstream service tool – **not** part of Bosch CDR system
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a pattern of aggressive driver behavior. Odometer included. Great for Road Rage cases.
- Access this whenever EDR data is either ***not available*** or ***insufficient*** for your investigation

New “13 EDR” & “15 EDR”

- **13 EDR** Adds Lateral ***Accel*** data from ACM in frontal events for 200ms
  (Fixes biggest drawback of **12 EDR**)

Longitudinal/Lateral Crash Pulse (Most Recent Event

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	F
0	0.0 [0.0]	0.0	
10	0.3 [0.6]	-1.9	
20	0.5 [0.8]	0.0	

- 10ms interval
- DV_{10ms} = ***Accel***/9.8*(32.2/1.466)*0.01s
- Add up intervals to get ***total*** **Delta V**

New “17 EDR” for Ped Protect

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM recording added to frontal events
- More precrash registers (6 vs 2 prior)
- More front/rear r& side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of satellite sensors

NISSAN EDR's 17.5 (CDR 34%)

Model	Location	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Last 3 digits of Bosch CDR cable number														
INFINITI														
G CONVERTIBLE	TUNNEL BETWEEN SEATS				*		Jul-12		780					
G COUPE	TUNNEL BETWEEN SEATS		*						780					
G SEDAN	TUNNEL BETWEEN SEATS						Jul-12		780					
M	TUNNEL BETWEEN SEATS								780					
Q40	TUNNEL BETWEEN SEATS										780			
Q50	TUNNEL BETWEEN SEATS								780	780	780	780	780	780
Q60	TUNNEL BETWEEN SEATS								780	780	780		780	780
Q70	TUNNEL BETWEEN SEATS								780	780	780	780	780	780
EX	TUNNEL BETWEEN SEATS								780					
FX	TUNNEL BETWEEN SEATS								780					
JX35	TUNNEL BETWEEN SEATS								780					
QX50-80	TUNNEL BETWEEN SEATS								780	780	780	780	780	QX60 so fa
QX30	TUNNEL BETWEEN SEATS												780	780
NISSAN		2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
370Z	TUNNEL BETWEEN SEATS		*				Aug-12		780	780	780	780	780	780
ALTIMA COUPE	TUNNEL BETWEEN SEATS			*					780					
ALTIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780
ARMADA	TUNNEL BETWEEN SEATS	*							780	780	780	?	780	780
CUBE	TUNNEL BETWEEN SEATS				*				780	780				
FRONTIER	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780	780	780	780
GT-R	TUNNEL BETWEEN SEATS				*					780	780	780	780	780
JUKE	TUNNEL BETWEEN SEATS								780	780	780	780	780	
LEAF	TUNNEL BETWEEN SEATS								780	780	780	780	780	
MAXIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780
MICRA	Check Manual										780	780	780	780
MURANO	TUNNEL BETWEEN SEATS		*		*				780	780	780	780	780	
MURANO CROSS C	TUNNEL BETWEEN SEATS								included in above					
NV200	TUNNEL BETWEEN SEATS								780	780	780	780	780	780
PATHFINDER	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780
QUEST	CTR CONSOLE NEAR FRONT		*						780	780	780	780	780	
ROGUE	Check Manual			*					780	780	780	780	780	780
SENTRA	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780
TITAN	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780
VERSA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780		780
VERSA NOTE HB	TUNNEL BETWEEN SEATS						Sep-12		780	780	780	780	780	780
XTERRA	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780			

Yellow = Proprietary Consult Tool Only

Yellow area not covered by CDR

Dealers do NOT have special software required to read EDR

No direct to module cables

563

Nissan Pre-Crash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % full	Engine RPM	Motor RPM	Service Brake (On, Off)	Steering Input (deg)
-5.0	24 [39]	16	2650	0	Off (Brake Not Activated)	-32
-4.5	23 [37]	11.5	1750	0	Off (Brake Not Activated)	-2
-4.0	21 [34]	9	1200	0	On (Brake Activated)	28
-3.5	15 [24]	9	1150	0	On (Brake Activated)	14
-3.0	13 [21]	4	950	0	On (Brake Activated)	4
-2.5	11 [17]	4	950	0	On (Brake Activated)	4
-2.0	5 [8]	4	900	0	On (Brake Activated)	-48
-1.5	3 [5]	4	850	0	On (Brake Activated)	-98
-1.0	1 [2]	4	850	0	On (Brake Activated)	-160
-0.5	3 [5]	10.5	1100	0	Off (Brake Not Activated)	-194
0.0	6 [9]	16	1150	0	Off (Brake Not Activated)	-218

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

Nissan Polarity Issues Fixed?

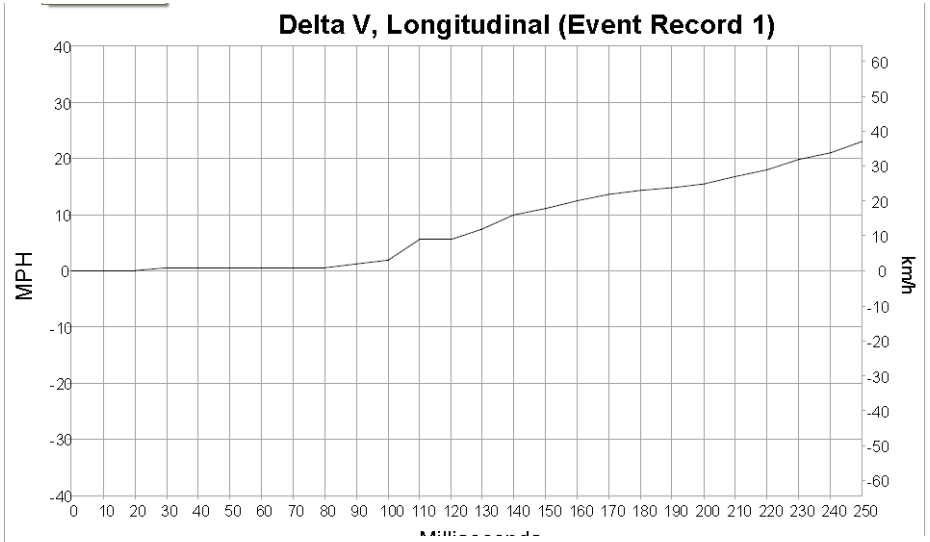
Some exceptions still reported

2013 Altima in CDR 12.2

System Status at Event (Event Record 1)

Frontal Air Bag Suppression Switch Status	S airbag inhibit)
Maximum Delta-V, Longitudinal (MPH [km/h])	29 [47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

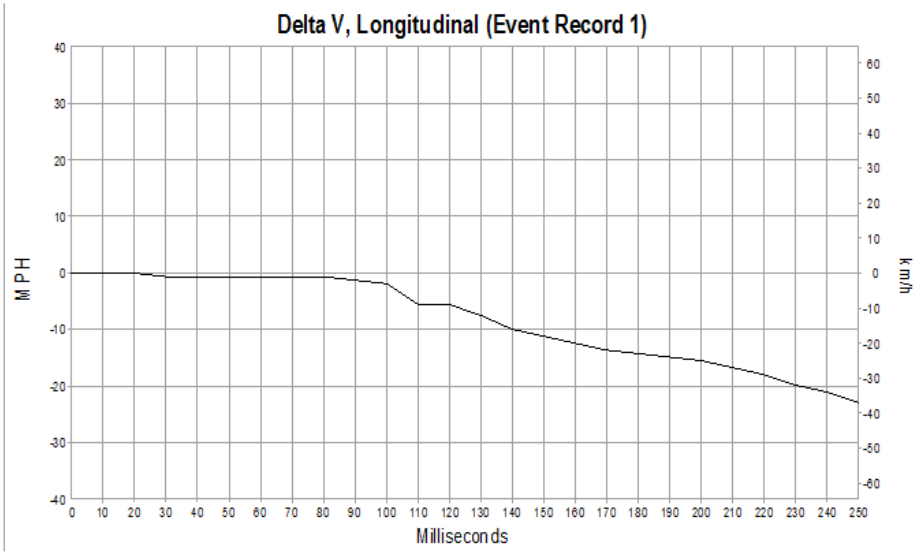
+29 at 300ms



2013 Altima in CDR 16.5

	S airbag inhibit)
Maximum Delta-V, Longitudinal (MPH [km/h])	-29 [-47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	-32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

-29 at 300ms



Mazda CDR 17.5 Coverage 34%

Model	Location										
Last 3 digits of Bosch CDR cable number shown		2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
MAZDA				All Mazda use 387 adapter for direct to module							
CX-3	TUNNEL BELOW IP								812		
Tribute/CX-5	TUNNEL AFT SHIFTER					779			811		
CX-9	TUNNEL BELOW IP			778							
Mazda 2	TUNNEL AFT SHIFTER			792				not listed	812	not listed	
Mazda 3	TUNNEL AFT SHIFTER				779				811		
Mazda 5	TUNNEL BELOW IP				778					not listed	
Mazda 6	TUNNEL BELOW IP			792			779		811		
MX-5 Miatta	TUNNEL BELOW IP					no coverage			812		

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers. Mostly Delta V.

Memo: The Mazda Tribute was sister to the Ford Escape thru 2011. A 2011 Tribute was read using a 2011 Escape VIN.

Typical Mazda Precrash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, % full	Service Brake (On, Off)
-5.0	19 [31]	0	On
-4.5	17 [27]	0	On
-4.0	14 [23]	0	On
-3.5	12 [19]	0	On
-3.0	11 [17]	25	Off
-2.5	10 [16]	61	Off
-2.0	11 [17]	63	Off
-1.5	13 [21]	64	Off
-1.0	16 [26]	64	Off
-0.5	17 [27]	0	Off
0.0	4 [6]	0	Off

Minimum Regulatory Content

Recording threshold <5mph

This is a second event – last data pt at impact

Algorithm woke up early, missed later event
Noted in newer Fords, Mazda & Kia/Hyundai



System Status at Event (Event Record 1)

Safety Belt Status, Driver	Belted
Safety Belt Status, Right Front Passenger	Unbelted
Occupant Size Classification, Front Passenger	Adult
Frontal Air Bag Warning Lamp (On, Off)	Off
Ignition Cycle, Crash	1892
Multi-Event, Number of Events (1, 2)	No. 1
Complete File Recorded (Yes/No)	Yes
Ignition Cycle, Download	1893
Maximum Delta-V, Longitudinal (MPH [km/h])	1.2 [2]
Time, Maximum Delta-V, Longitudinal (msec)	297.5
Maximum Delta-V, Lateral (MPH [km/h])	-1.2 [-2]
Time, Maximum Delta-V, Lateral (msec)	297.5
Time, Maximum Delta-V, Resultant (msec)	297.5

Deployment Command Data (Event Record 1)

Pretensioner Deployment, Time to Fire, Driver (msec)	250
Pretensioner Deployment, Time to Fire, Right Front Passenger (msec)	250
Frontal Air Bag Deployment, Time to Deploy/First Stage, Driver (msec)	250
Frontal Air Bag Deployment, Time to Deploy/First Stage, Right Front Passenger (msec)	250

Max Value
Mazda

Volvo CDR 17.5 Coverage 17%

Model	Location	2010	2011	2012	2013	2014	2015	2016	2017
Last 3 digits of Bosch CDR cable number sho									
VOLVO		All Volvo use 387 Adapter for DTM images							
S60	CTR TUNNEL UNDER CONSOLE		799						
S80, S80L	CTR TUNNEL UNDER CONSOLE								
S90	CTR TUNNEL UNDER CONSOLE								816
V40	CTR TUNNEL UNDER CONSOLE								816 Euro
V60	CTR TUNNEL UNDER CONSOLE		799						
V70	CTR TUNNEL UNDER CONSOLE				799				
V90	CTR TUNNEL UNDER CONSOLE								816
XC60	CTR TUNNEL UNDER CONSOLE				799				
XC70	CTR TUNNEL UNDER CONSOLE				799				
XC90	CTR TUNNEL UNDER CONSOLE							816	

Old Volvo DV only Law Enf only call Volvo

The source data for this chart is the Bosch CDR help file for Version 17.

Always consult the latest help file for the most accurate coverage

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Volvo Sample Pre-Crash Data

Pre-Crash -5 to 0 sec (Event Record 1)

Mostly minimum to meet 563

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, Percent Full (%)	Service Brake (On, Off)
-5.0	3.1 [5.0]	0.0	Off
-4.5	3.1 [5.0]	0.0	Off
-4.0	3.1 [5.0]	0.0	Off
-3.5	5.0 [8.0]	1.0	On
-3.0	8.1 [13.0]	3.0	Off
-2.5	11.2 [18.0]	5.0	Off
-2.0	14.3 [23.0]	7.0	Off
-1.5	17.4 [28.0]	9.0	On
-1.0	20.5 [33.0]	11.0	On
-0.5	23.6 [38.0]	13.0	On
0.0	26.7 [43.0]	15.0	Off

No RPM

No Steering

Some reports of last data point taken after AE received

BMW CDR 17.5 Coverage 28%

See help file for Rolls Royce

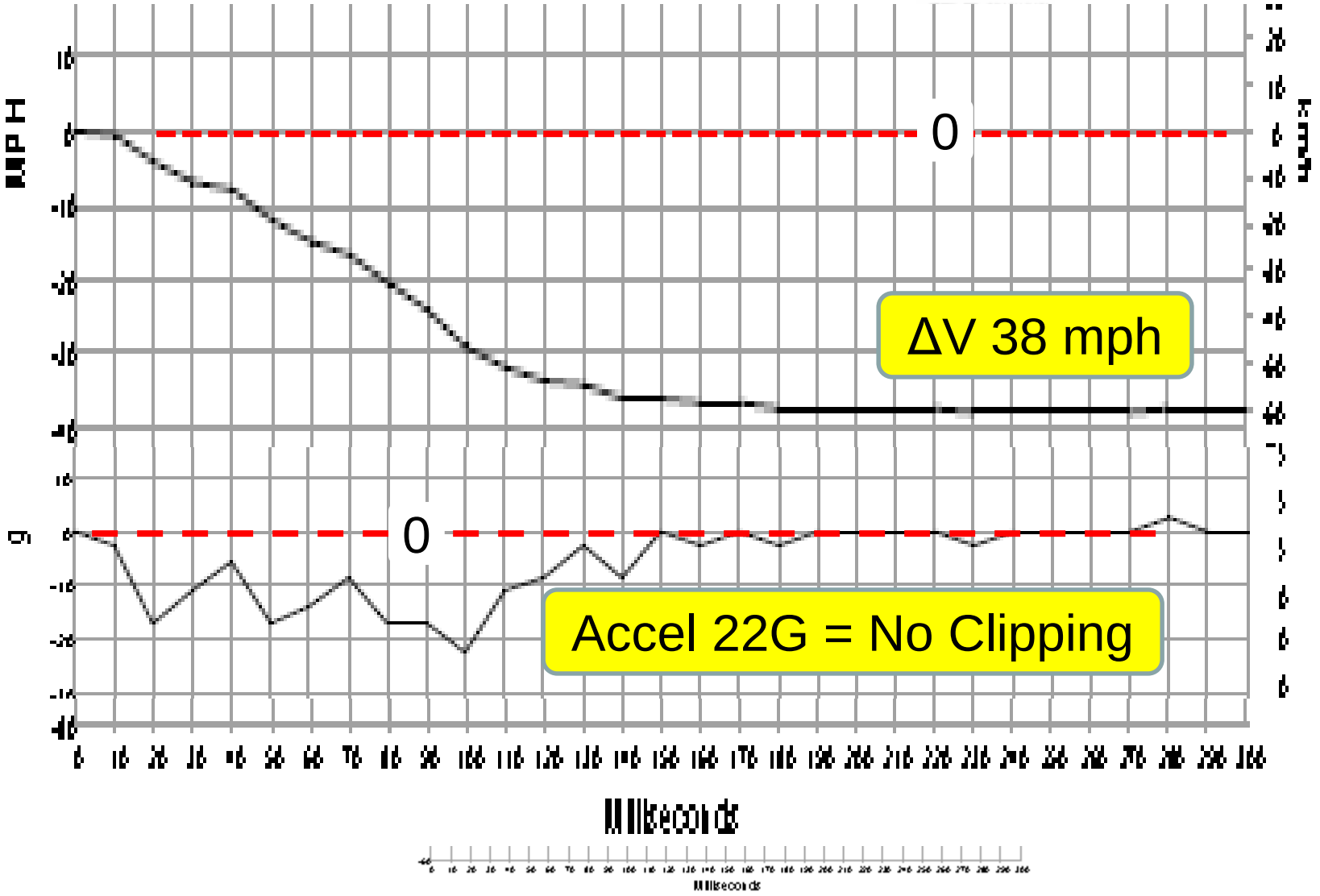
Model	Location	2012	2013	2014	2015	2016	2017	2018
Last 3 digits of Bosch CDR adapter/cable number shown								
BMW	**All BMW require 387 Adapter for DTM**							
1 Series					not covered - Europe Only			
2 Series	Driver side, behind IP, rt of strg			796				
3 Series	Driver side, behind IP, rt of strg		796					
Active Hybrid 3	Passenger side, behind the glove box		796					
4 Series	Driver side, behind IP, rt of strg			796				
5 Series	Passenger side, behind the glove box		796					500/822
6 Series	Passenger side, behind the glove box		796					
7 Series	Passenger side, behind the glove box		796			500/796	500/822	
X3 xDrive	Passenger side, behind the glove box		796					
X5	Center Tunnel, Between Seats			796				
X1	Center Tunnel, Between Seats				798	500/798	500/807	
X3, X4, X6	See Help File - Various				796			
Z4	Center Tunnel, Under Dash				798			
i3, i8	See Service Manual			500/807				

BMW Sample Pre-Crash

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Engine RPM	Steering Input (deg)	Service Brake, On/Off	ABS Activity (Engaged, Non-engaged)	Stability Control (On Engaged, Non-engaged)
-5.0	55 [89]	16	1500	0	Off	No ABS Activity	Non-engaged
-4.5	55 [89]	17	1500	0	Off	No ABS Activity	Non-engaged
-4.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.5	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-2.5	55 [89]	8	1500	0	Off	No ABS Activity	Non-engaged
-2.0	55 [89]	7	1500	0	Off	No ABS Activity	Non-engaged
-1.5	55 [89]	14	1500	0	Off	No ABS Activity	Non-engaged
-1.0	55 [88]	19	1500	0	Off	No ABS Activity	Non-engaged
-0.5	55 [89]	0	1500	0	Off	No ABS Activity	Non-engaged
0.0	45 [73]	0	1300	10	On	ABS Activity	Non-engaged

BMW – Both ΔV & Accel



Recent BMW news

- Many updates to software – be SURE to use latest software to read and interpret – DLC was not working on some models due to gateway module issue
- Lateral DV polarity MAY be inverted on 2014 Mini, BMW i3/i3rex, i8 – if car was serviced after 3/2015 it may be fixed.
- OCS “features” under investigation
- BMW owns Rolls Royce, anyone gets a crash on one, call instructors immediately

BMW EDR Evolution

- Basic EDR 2013
- “Extension” 2014
- Some driver assist mid-2015 - timing
- Auto Steering (autonomous vehicle)

  Cameras all views 4 fps

  Not intended for CDR,

  only BMW can access

Mercedes 17.5 21%

Model Last 3 digits of Bosch CDR DTM	Start of Sales cable number	2013	2014	2015	2016	2017	2018
Mercedes	EDR Start	All Mercedes use 387 adapter for direct to module downloads					
AMG-GT					800		
B-Class EV	after 4/1/2014			598			
B-Class	after 6/2/2014			598			
C Class Sedan	after 9/1/2014			801			
C Class Coupe	start of 2015 M.Y.			800	801		
CLA Class	after 6/2/2014			598			
CLS-Class				800			
E Class (includes coupe)	after 4/1/2013		800				801
G Class	after 11/1/2014			598			
GL Class	after 7/11/2014			598	?	?	
GLA Class	after 6/2/2014			598			
GLC Class					598		801
GLE Class					598		
GLK	after 12/2013		800		?	?	
GLS Class					598		
Maybach S600					801		
ML-Class	after 7/11/2014			598	?	?	?
S-class Sedan	after 9/2013		801				
S-Class Coupe	after 9/1/2014			801			
SL	after 12/2013		800				
SLK	after12/2013		800				SLC 800
Vito					598		
Smart & 2016 with "E"	after 9/1/2014			809			
Smart with VIN 4th "F"					820		
Smart for two							820
Metris							598

The source data for this chart is the Bosch CDR help file for Version 17.5 and Mercedes personal

Mercedes Details

- Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
- Precrash speed, brake, throttle & longitudinal ΔV, 563 table 1 items only.
- Can hold 5 events, D's lock, ND's FIFO
- Recording threshold is 5 mph over 150ms
- Speed is avg. of 4 wheel speed sensors

Any EDR data in older Mercedes? NO

- Some diagnostic codes may include “freeze frame” data, but airbag codes don’t record speed. Some codes do.
- By the time those other codes get set after a crash, the speed is no longer the precrash speed.
- Mercedes will assist law enforcement. Call 1-88-4Mercedes.

VW/Audi

- VW/Audi are the 13th and 14th brands to join Bosch CDR.
- VW/Audi are just phasing in EDR's starting with 2014 EOS, most models get EDR in 2015 model year - which was based on the Dec 2012 NPRM would have required EDR for Sept. 2014.

VW/Audi CDR 17.5 Coverage

See help file for Rolls Royce

Model	Location	2009-13	2014	2015	2016	2017	2018
Last 3 digits of Bosch CDR adapter/cable number							
All VW/Audi Require the 387 Adapter to read DTM							
VW							
Atlas	UNDER CENTER STACK						805
Beetle	UNDER CENTER STACK			804			
CC	UNDER CENTER STACK			804			
EOS	UNDER CENTER STACK		806				
Golf, eGolf	UNDER CENTER STACK			805			
Jetta	UNDER CENTER STACK			804			
Passat	UNDER CENTER STACK			804			
Routan	UNDER CENTER STACK	785					
Sportwagen				not covered at this time			
Tiguan	UNDER CENTER STACK			804			
Touareg	UNDER GEAR SELECTOR			804			
AUDI				2015	2016	2017	2018
A3,S3	UNDER CTR STACK			805			
A4, allroad	Tunnel between 1st/2nd row seats			804		500/813	
A5, S5, RS5	Tunnel between 1st/2nd row seats			804			500/813
A6, S6	Tunnel between 1st/2nd row seats			804			
A7, S7, RS7	Tunnel between 1st/2nd row seats			804			
A8, S8	Tunnel between 1st/2nd row seats			804			
Q3	UNDER CTR STACK			804			
Q5, SQ5	Tunnel between 1st/2nd row seats			804			500/813
Q7	Under Center Stack					500/813	
R8	Under Center Stack					804	
TT, TTS				not covered at this time			

Chrysler Minivan Sister

No “back model” coverage

Other VW Brands & Info

- **No EDR** in Porsche yet- but it is only 0.3% of the US market
- 4 Bentley models 2016-2017 now covered - 0.0% of US market
- Lamborghini Adventadore & Hurracan covered 2015-2016 (**not** 2017 yet?)

VW/Audi Details

- Same data set for all event types
- Time from last precrash to AE noted
- Recorder holds up to 6 ND events in Chronological order.
- New modules will use Flex Ray
- Date/Time of Crash from clock (GPS if clock not set) in some models (Audi?)

VW/Audi Details

- Modules live for 15 sec after power out
- Image time 1-5 minutes
- Pushbutton ignition – only 1 push to ACC
- Battery disconnect after serious crashes may force backpowering
- Must power ACM **and** gateway module

TESLA

- Tesla records MANY data parameters every one second plus when anything electronically monitored happens.
- It is NOT an “event” data recorder, it is a continuous recorder (does **NOT** meet Part 563 definition of event recorder) (so **NO** PUBLICLY AVAILABLE tool to read it)
- Law Enforcement contact Tesla and demonstrate proper legal authority
- Tell them what you are interested in

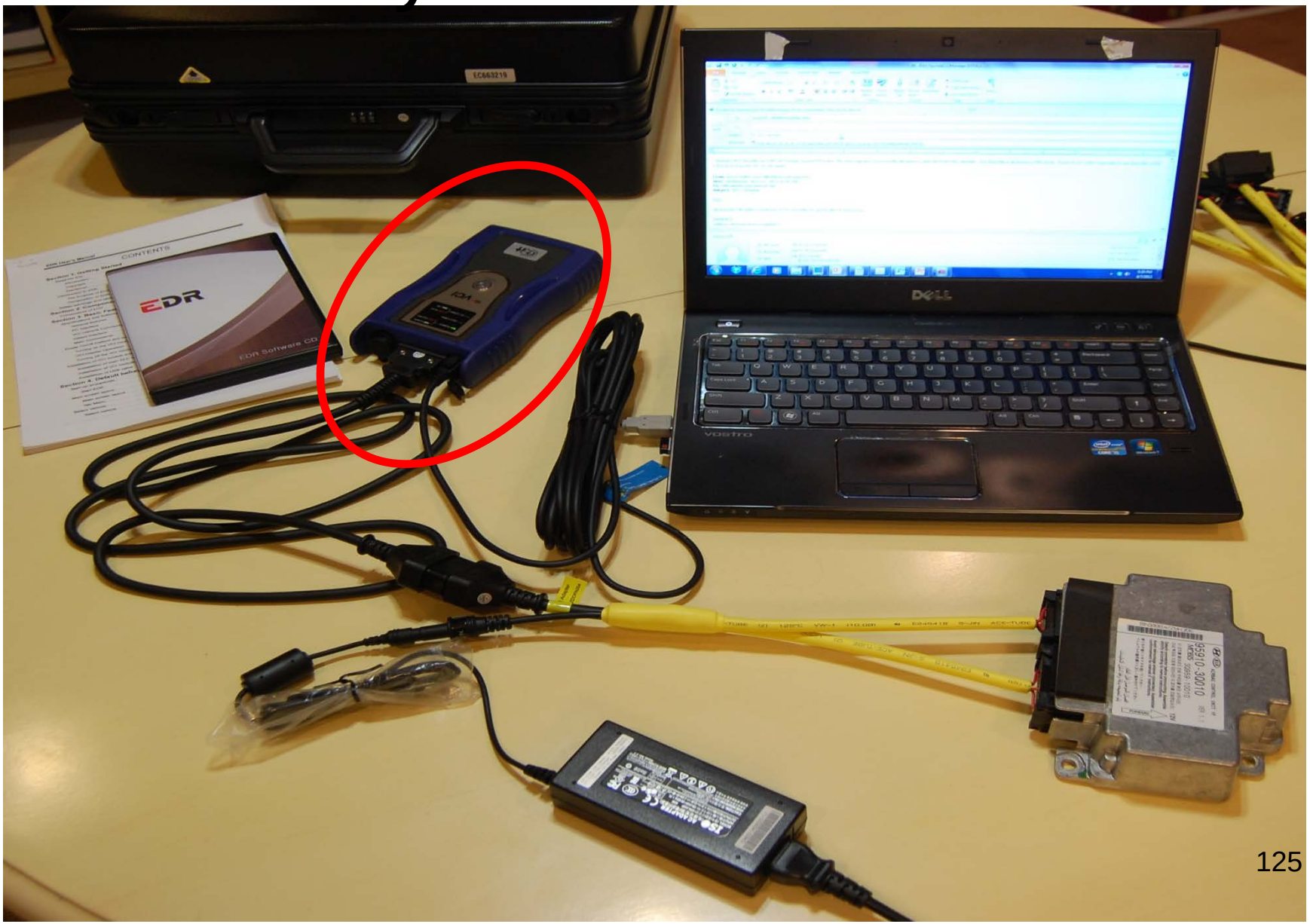
Hyundai/Kia Tools

- There is no DLC only kit. Hyundai kit comes with 12 DTM cables, Kia with 15 DTM cables. New cables cost $300.
- Hyundai: Call GIT America Tool Company (714)433-2180 order Part G0Z**H**DMN001 ***$4950***
- Kia" Call Dealer Tools hotline 888-542-1011, order part GIT0Z**K**DMN001 ***$7098 + tax+ship www.kiaspecialtools.com***
- Software updates $295/yr per brand from GIT Tool Company. Now at software V23.

Additional Info

- K/H Interface modules are identical, but you must buy both to get software for both. Software updates won't download unless the correct interface module is attached to your laptop.
- I read K/H modules for others for $500.
- Many more people are buying these kits now that we have 5 model years on the road

The Hyundai EDR Reader



Hyundai/Kia Tool Info (not new)

- The tool “officially” only reads 2013 and later models – software requires VIN input and locks out 2010-2012.
- But it communicates with several 2010-2012 models.
- However, your report will say you told the software it is a 2013.
- Admissibility must be presented skillfully (get help). **Data successfully admitted in Virginia and California.**

Kia/Hyundai GIT Tool May 2017

		Model Years							
Model	Vehicle Size, Descrip	2010	2011	2012	2013*	2014	2015	2016	2017
KIA		Cable #	Cable #	Cable #	Cable #	Cable #	Cable #	Cable #	Cable #
CADENZA (VG, 2014) (Azera sister)	112.0" WB	YG requires cable 13 per software				19(VG)	19	19	13(YG)
FORTE (TD 2013, YD 2014))	104.3"WB, compact	10	10	10	10 (TD)	6/12 (YD)	6/12 (YD)	6/12 (YD)	6/12 (YD)
K900 (KH 2015)	119.9"WB						11 (KH)	11	11
NIRO									21(DE HEV)
OPTIMA (2013 TF QF, 2016 JF, JFA)	110.0" WB sedan	No	4	4	4 (TF/QF)	4	4	14 (JF,JFA)*	14
OPTIMA HYBRID	110.0" WB sedan	No	4	4	(TF HEV)	4	4	4	23(JFHEV)
RIO (UB)	94.5"WB, compact	No	No	8	8(UB)	8	8	8	8
RONDO (Canada only 2011 MY+)		No	out of production						
SEDONA (VQ '14, YP '15)	113.8/118.9" WB miniv	No	No	No	No	10(VQ)	12(YP)	12	12
SORENTO (XM to 2015)(UMA 2016)	106.3" WB, crossover	No	3	3	3(XM)	7(XM)	7	13 (UMA)	13
SOUL (AM 2013, PS 2014+)	100.4" WB	1 see note 1	1 see note 1	1	1(AM)	12(PS)	12(PSEV)	12	12
SPORTAGE (SL)	103.9" WB, crossover	No	Yes	2	2(SL)	2	2	2	13(QL)
HYUNDAI									
ACCENT (RB)	94.5"WB, compact	No 10	No?	5 see note 2	5(RB)	5	5	5	5
AZERA (HG)	109.4"WB, sedan		likely	Yes	7(HG)	7	7	7	7
ELANTRA Coupe(JK), Sedan(UD/MD)	104.3"WB, compact	No	Yes	8	8(JK/UD/MD)	8	8	8	21(AD, ADA)
ELANTRA GT (GD) (aka "i30")	former Elantra Touring	old model		new model	6(GD)	6	12	12	12
EQUUS (VI)(stretch genesis)	119.9" WB, luxury seda	?	yes?	yes?	9(VI)	11(VI)	11	11	
GENESIS (Coupe BK, Sedan BH/DH))	115.6" WB, luxury seda	9 shave keyway	9 shave keyway	?	9(BK, BH)	9	13(DH)	13	13
GENESIS G90									13(HI)
SANTE FE (Sport DMA, NC)	106.3" WB, crossover	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)	7	7	7	7
SONATA (YF sedan, YF HEV)	110.0" WB sedan		4	?	4(YF)	4	14 (LF) 4 (HEV)	14 (LF HEV)	14
TUCSON (LM to 2015)(TL in 2016)	103.9" WB, crossover	Yes	Yes	2	2(LM)	2	2	13(TL)	13
VELOSTER coupe (FS)	104.3"WB, compact	out of productior		?	5(FS)	5	5	5	5
Gold = 2013 Part 582 Compliant	Pink = Some data may be inaccu	K4 Common Cables							

New GENESIS brand

- The Luxury brand of Hyundai
- G80 and G90 models
- Read with Hyundai data recorder tool

Kia/Hyundai Tool Issues

- You input VIN, it prints what you entered in report. VIN is not stored in the ACM. Do sufficient documentation to prove your readout came from your vehicle.
- The K/H report is a PDF file. There is no raw data file. If K/H ever discovers an error and updates software, you will have to re-read the module. The most recent release now updates software version in the report
- Recommended: ***Secure module*** as evidence

K/H 2010-2012 data reliability?

- In 50+ NHTSA crashes, speed, delta V and belt use appear OK (one exception)
- Steering angle, and accel data appear implausible on some
- Key cycles at download not in all models
- SAE 2014-01-0502 has details.
- Additional testing on 2012 Soul reported in SAE 2015-01-1445.

Software Upgrade April 2016

- Auto-detects VIN in DLC readouts. Requires manual VIN entry for DTM readouts.
- IF reading 2010-2012 models, you must TURN OFF THE KEY during VIN autodetect until it fails.
- Then you must enter a surrogate 2013 VIN.
- Take your VIN, change 10th digit to a D, change last 5 to all zeros, iterate 9th check digit. You can use CDR VIN entry to validate when you found the right check digit.
- Some user report software hangs on pre-2013's, you must load original V1 software and use that.

Other Non-Bosch CDR Tools

Law Enforcement: Ask the dealer to help you

	2012 Nov YTD SALES	Nov YTD US Market SHARE	2013 EDR?	Tool?
SUBARU	299,788	2.3%	YES	Hitachi HDS-3000
Dealer scan tool $2835 plus software $1835/mfr (905)669-4812				
New Denso tool from Nuspire required for 2016+, $2K + $5800 software/yr				
MITSUBISHI	53,677	0.4%	YES	OTC tool co
Panasonic Touchpad + software $6395, annual $1200/yr, 888 727 6672				
JAG/LAND ROVER (Tata Motors India)	49,452	0.4%	YES	OTC/Bosch SPX
Full dealer scan tool, Toughbook & software approx $5000				
SAAB	480	0.0%	YES?	Hitachi HDS-3000?
Same hardware as Subaru, software $1835 for Saab				

Hitachi KDS-3000 Interface

Subaru Tool Coverage (not CDR)

Denso DSTi Interface

SSM3 Software

SSM4

Model	2011	2012	2013	2014	2015	2016	2017
SUBARU		<-------Old Hitachi tool or new Denso Tool----->				New Denso Tool	
BRZ	n/a	n/a	Yes (Scion FRS sister)	Yes	Yes	Yes	Yes
Impreza	No	Yes	Yes	Yes	Yes	Yes	Yes
WRX/STI	No	No	No	No	Yes	Yes	Yes
Legacy	No	No	Yes	Yes	Yes	Yes	Yes
Outback	No	No	Yes	Yes	Yes	Yes	Yes
Forester	No	No	No	Yes	Yes	Yes	Yes
Tribeca	No	No	No	No	Discontinued		
Crosstrek	n/a	n/a	Yes	Yes	Yes	Yes	Yes

NOTE: The Subaru tool is a dealership fault code tester made to work in thru the DLC in vehicles with functioning electrical systems only.

Instructors may be able to fabricate a direct to module cable from a vehicle side connector

Subaru EDR Tool

- Subaru used the Blue Streak Electronics “Subaru Diagnostic Interface” hardware with “Subaru Select Monitor III” software from its first EDR’s in 2012 thru 2015. This is approximately $3K for hardware and $2K for software annual subscription. SSM**3**
- For 2016 Subaru has gone to a new Denso DSTi diagnostic interface with new SSM**4** software from Nuspire. Hardware $2K software $5.8K/yr, is compatible to ‘04.

Subaru

- The Dealer tool is DLC only. The electrical system must be working, or you must backpower both ACM and DLC fuses, or put ACM in surrogate vehicle.
- Tutorial available on software use
- Breaking news – 2015 tool may still work on some 2016 and 2017 models!
- Breaking news – 2016+ SSM4 software does not have a PDF or CSV output – must be viewed in software

Subaru DTM Cable Available

ATTENTION:

GENERAL MANAGER ❑
PARTS MANAGER ❑
CLAIMS PERSONNEL ❑
SERVICE MANAGER ❑

IMPORTANT - All Service Personnel Should Read and Initial in the boxes provided, right.

SUBARU

QUALITY DRIVEN® SERVICE

SERVICE BULLETIN

APPLICABILITY: 2013MY and Later Legacy and Outback
2014MY and Later Forester
2012MY and Later Impreza 2.0L NA
2013MY and Later XV Crosstrek
2013MY and Later BRZ
2015MY and Later WRX and WRX STI

NUMBER: 17-17-13R
DATE: 03/18/13
REVISED: 05/20/14

SUBJECT: Event Data Recorder (EDR) Function of Airbag Module

RUMOR that CDR may support Subaru in 2019??

INTRODUCTION

NOTE: The Airbag systems used on previous and current WRX, STI and Tribeca models are NOT equipped with EDR functionality.

PARTS INFORMATION

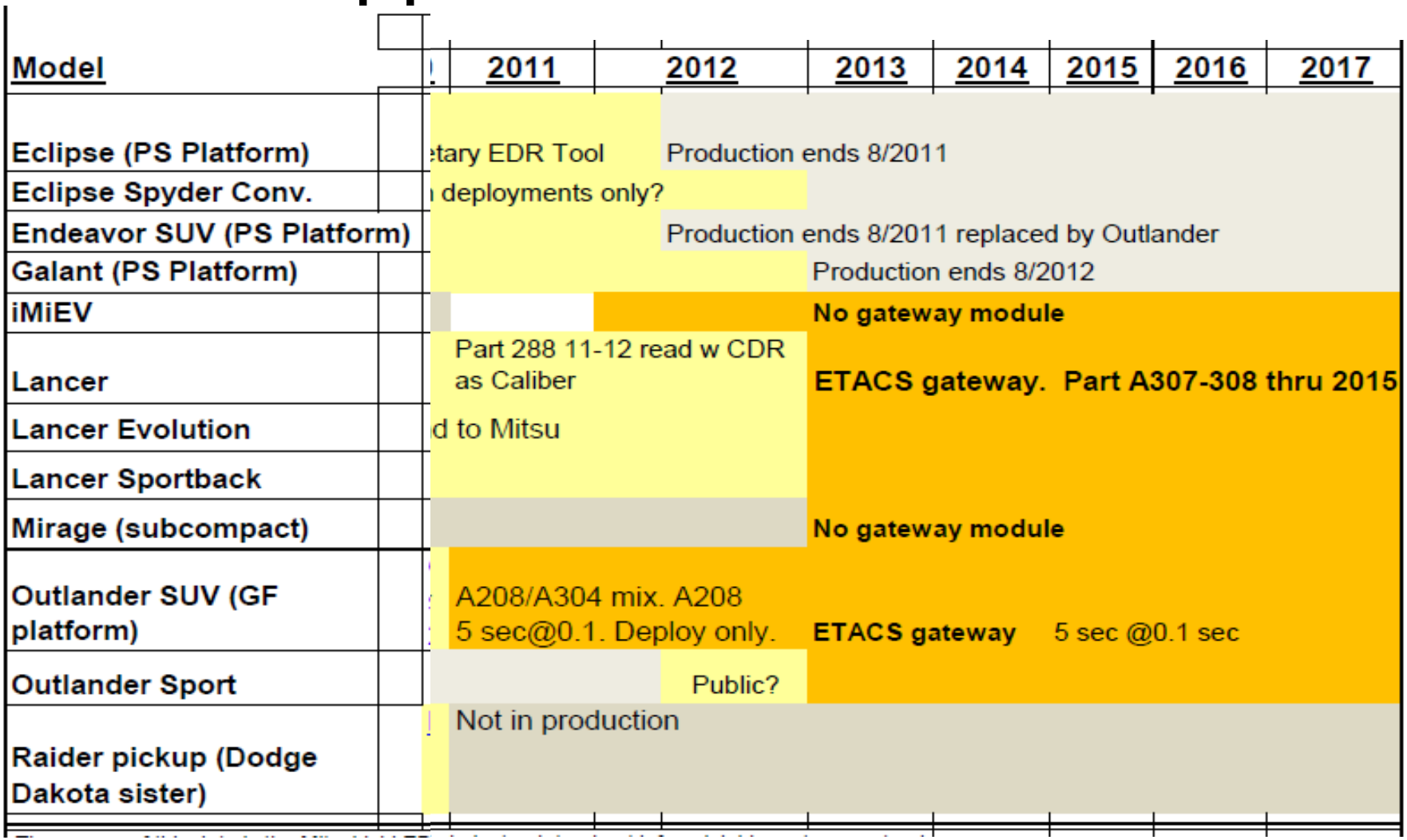
The test harness required to access the EDR data from the airbag control module along with current pricing is available from SPX / Kent-Moore (1-800-345-2233). The harness part number is 98299FJ040.

Dealer can't buy it, cost is $1060, must talk to God to order

MITSUBISHI 0.5% market share

- Mitsubishi EDR only tool based on their dealership a Panasonic Touchpad. Cost to $6380 + tax.
- The kit is made to work only on vehicles with a functioning electrical system thru the **DLC**.
- Some Mitsubishi have an **“ETACS” gateway module (interior fuse box)**that must be included in circuit to make it work DTM.
- Some incomplete files retrieved.
- Latest software possible bug – won’t convert .csv file into report format

Mitsubishi EDR's- Public Tool supports some 2011+



Mitsubishi EDR's- pre 2013

Model	2007	2008	2009	2010	2011	2012
Eclipse (PS Platform)		Part MU141170 4/2005 to 8/2011		Proprietary EDR Tool		Production e
Eclipse Spyder Conv.			?	Data in deployments only?		
Endeavor SUV (PS Platform)			?			Production e
Galant (PS Platform)			?			
iMiEV			Not sold in US			
Lancer			2.3 sec@0.1 deploys only		Part 288 11-12 read w CDR as Caliber	
Lancer Evolution			?	or send to Mitsu		
Lancer Sportback			?			
Mirage (subcompact)			Not in production			
Outlander SUV (GF platform)	Gen 3 starts in 2012		A208 Read w/CDR as Caliber		A208/A304 mix. A208 5 sec@0.1. Deploy only.	
Outlander Sport						Public?
Raider pickup (Dodge Dakota sister)	Bosch CDR supported cable 228				Not in production	

Jaguar/Land Rover

- Jag/LR have EDR's, they are read with a dedicated laptop and software. Tool only retrieves hex data, must be sent to England for interp. NOBODY HAS TOOL.
- Jag/LR have now designated 2 engineers in New Jersey HQ as EDR contacts.
- **They are now referring requesters to <u>suppliers.</u>** (call to find out who supplier is)
- Some Mid 2000's have ***PCM data*** (when Jag/LR was owned by Ford) but *not spoofable* with CDR.

Newly published research

- Every year at the SAE World Congress in April new research is presented.
- A very brief summary follows, I encourage you to read these papers.
- Other good sources of EDR info include Collision Magazine and Accident Reconstruction Journal.

2017 New EDR Research

SAE 2017-01-1436 Analysis of **EDR** Data in **Kawasaki** Ninja 300 **Motorcycles,** Fatzinger, Momenentum Eng.

(records if bike falls over while moving)

SAE 2017-01-1437 Intro to Forensic Acquisition of Pass Veh Infotainment & telematics systems (using Berla iVe), **Bortles** et al, Kineticorp

(very good for criminal investigations, limited value to reconstruction. Data not present in every circumstance)

2017 New EDR Research

SAE 2017-01-1438 Behavior of **Toyota** Event Data Recorders subjected to low and mid severity pulses – Lee, MEA

Continues prior documenting under reporting in some older ACM's

SAE 2017-01-1439 Commercial Vehicle Global Positioning Systems Based telematics data characteristics and limitations – Steiner, Mechanica (accuracy limits)

Technical Keynote: NTSB's Jane Foster "Heavy Vehicle Event Data recorders Needed to Improve Highway Safety".

EDR Regulatory Update
(no 2016-17 changes)

- Part 563 standardized minimum content for vehicles voluntarily equipped with EDR, became effective Sept 2012.
- NHTSA required 10% accuracy on Delta V Later clarified, 10% within range of sensor, requires OEM to report WHEN sensor first exceeds range (clips), effective 9/1/2014.
- Accuracy can be worse than +/-10% after clipping occurs

Regulatory Update

- NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (past!)
- NPRM also proposed EDR be made an FMVSS 405 (Safety Standard), OEM's would have to self certify EDR's. Strong OEM objections were filed.
- Still "Pending" as of this printing, once published followed by 60 day comment period before becoming effective **(date keeps slipping every time I check)**.

Regulatory Update

- US Senate Bill 1925 "**Driver Privacy Act**" was introduced 2014 with bi-partisan support, to **require warrant or owner consent** prior to accessing EDR. Passed as part of highway funding bill, signed into law Dec. 4, 2015.
- Requires NHTSA to **study how much data is really needed** & report to congress. No visible progress reported to my knowlege.

Regulatory Update

- NHTSA/IIHS got 10 auto mfrs to agree on 9/11/2015 to making **A**uto **E**mergency **B**raking **(AEB)** standard (timeline TBD). Voluntary commitment is a relatively new strategy by NHTSA. Now 20 have agreed.
- Next big issue is Autonomous Vehicles (self driving cars).
- States are drafting conflicting regulations on type of certification required – could hinder development.

EDR admissibility update

- “Every year EDR evidence is admitted at the trial level in hundreds of cases and we never hear about them.” - Prosecutor Bill Melkonian at the 2014 CDR User’s Conference Legal Update. **No change.**
- Warrants now required, admissibility shifts to Probable Cause for getting that warrant.
- Smaller battles on 2010-2012 Kia/Hyundai, Nissan 2006-2012, unsupported Ford PCM’s. etc.

EDR read before Privacy Act?

- Florida vs Worsham recent 2016 decision holds that warrant was still required before Driver Privacy Act. Court likened EDR to cell phones or tablets (we disagree). Went to appellate level.
- Worsham circumstances were unusual, police read module without warrant then applied for one. Judge denied the warrant since readout was already complete.

Admissibility Emerging Trends

- Kia/Hyundai Pre-2013 data
  - 2012 Kia Soul data admitted in CA criminal case – Vandiver literally ran a car crash to validate the data.
  - 2011 Hyundai Sonata admitted in VA v Chavis 4/7/2016 (Ruth), upheld 7/18/2017
- Unsupported Ford PCM data
  - “VIN spoofed” data passed Frye hearing in Colorado Criminal case March 2016 (Ruth)
- Mitsubishi incomplete file recorded admitted in CA (Vandiver)

Applying Automotive EDR Data to Traffic Crash Reconstruction

Introduction - END

Presented by Rick Ruth
313 910 5809
ruthconsulting@comcast.net



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